



Multi-slice spatial data integration

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Summary

1. Repeated measurement (uni-modal / multi-modal)

2. Different types of alignment

- Vertical alignment
- Horizontal alignment
- Diagonal alignment

3. Multi-modal dataset application

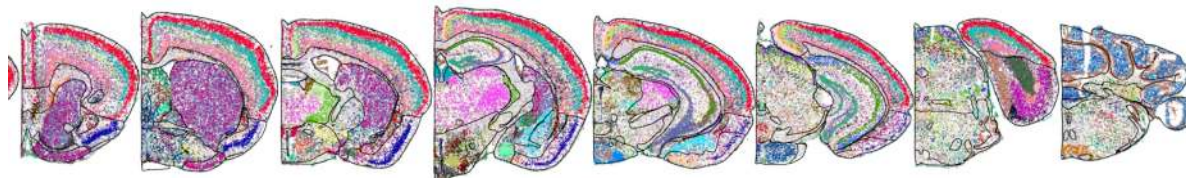
Why perform measurement on multiple sections?

Uni-modal consecutive sections

Same features measured

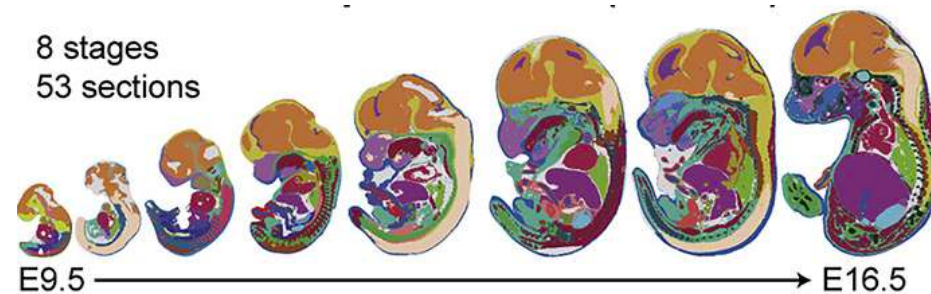
Replicates / atlas building

e.g. Spatial Transcriptomics



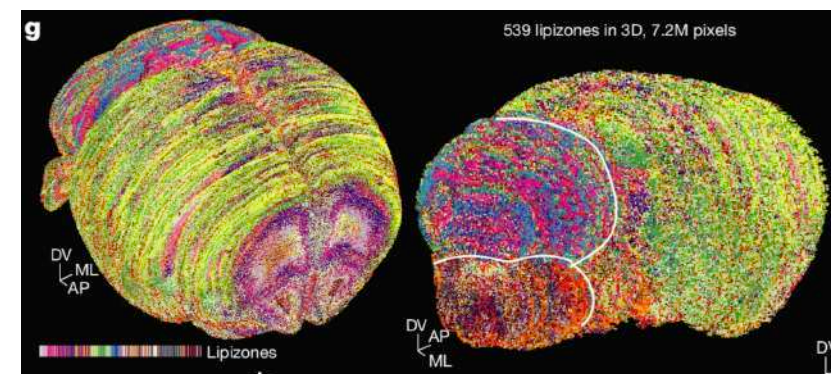
From Zhang et al. 2023 (Nature)

Developmental tissue atlases



From Chen et al. 2022 (Cell)

e.g. Spatial Lipidomics



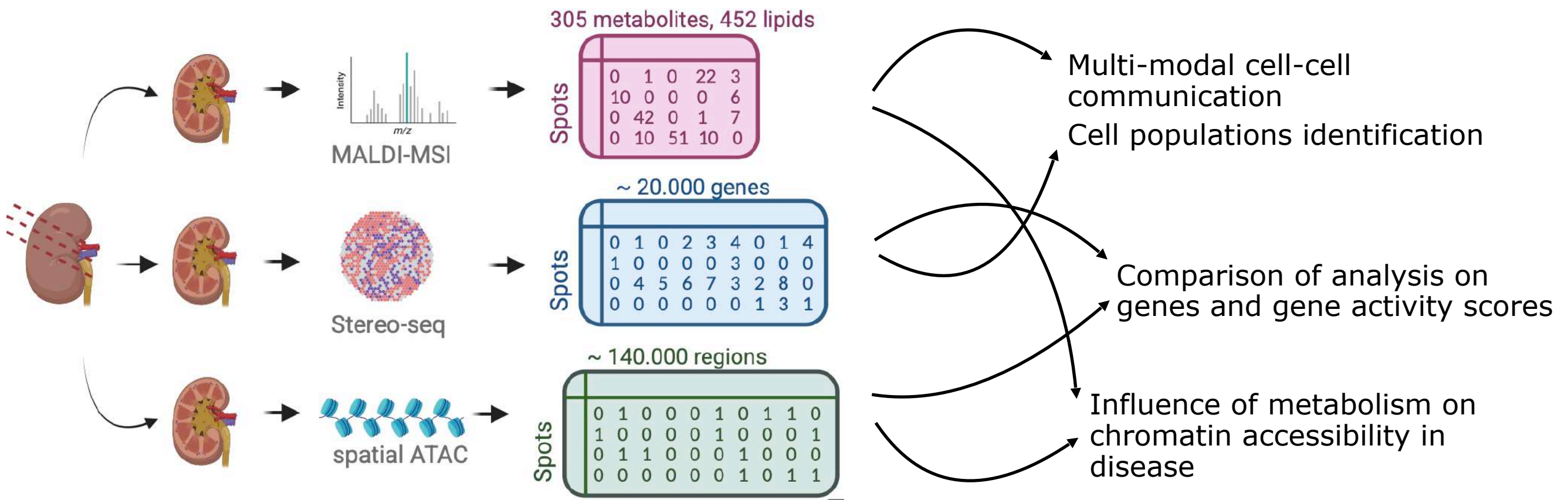
From Fusar-Bassini et al. 2026 (Nature)

Why perform measurement on multiple sections?

Multi-modal consecutive sections

Different molecular features measured
(= multi-modalities / multi-omics)

Different molecular layers = different views of the same tissue



Why align multiple tissue sections?

Without alignment

- Two separate datasets from the same tissue
- No matched observations between slices
- Spatial information cannot be combined across modalities
- Each dataset analysed in isolation
- Full multi-modal potential of the experiment is wasted



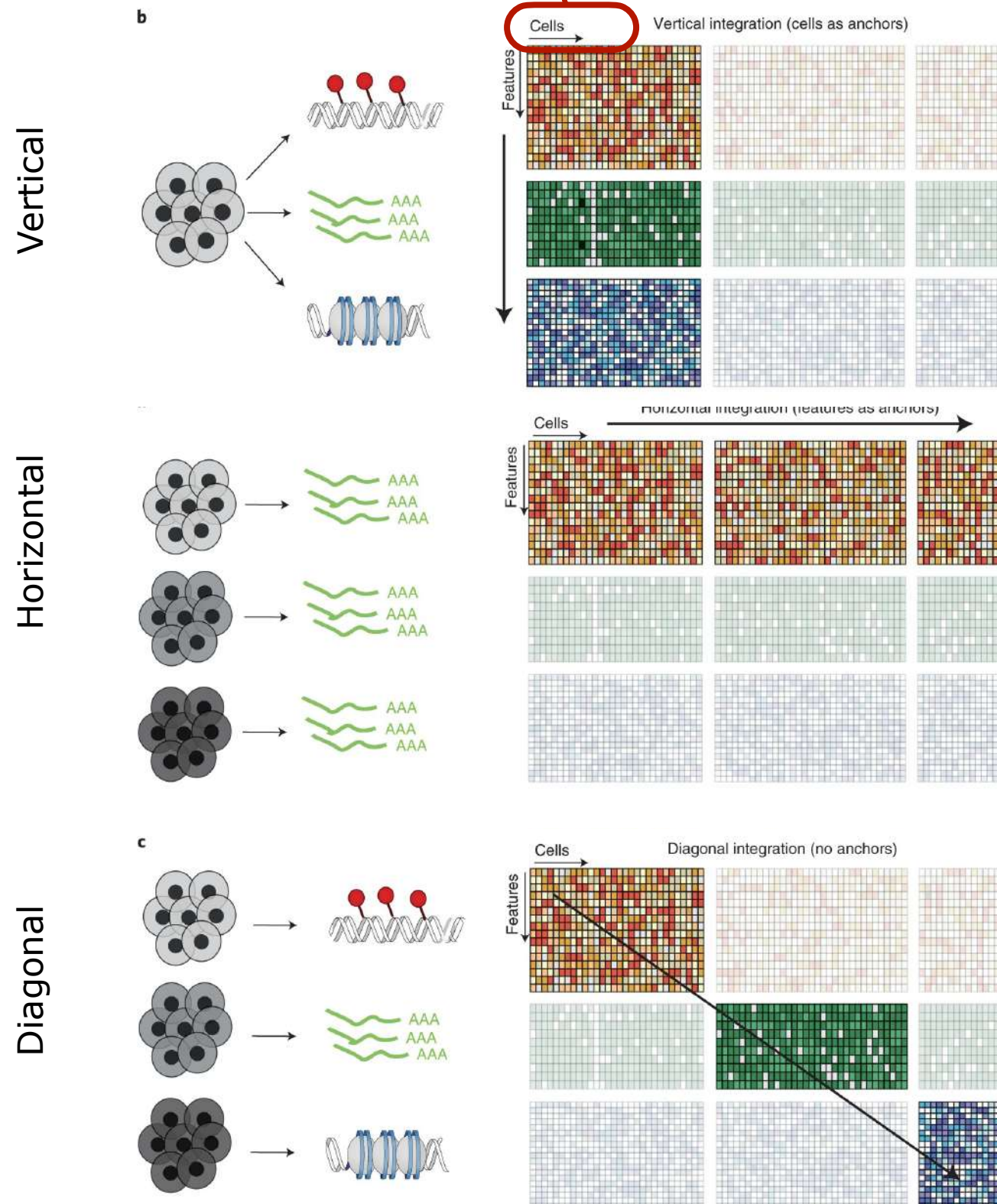
After alignment

- Matched observations across slices and modalities
- If multimodal: joint analysis of gene expression, chromatin, metabolites...
- Enables atlas building across individuals and experiments
- Full spatial context preserved and exploited

Three integration cases

In this spatial omics, observations are cells or spots with x,y coordinates

Increasing co-registration difficulty



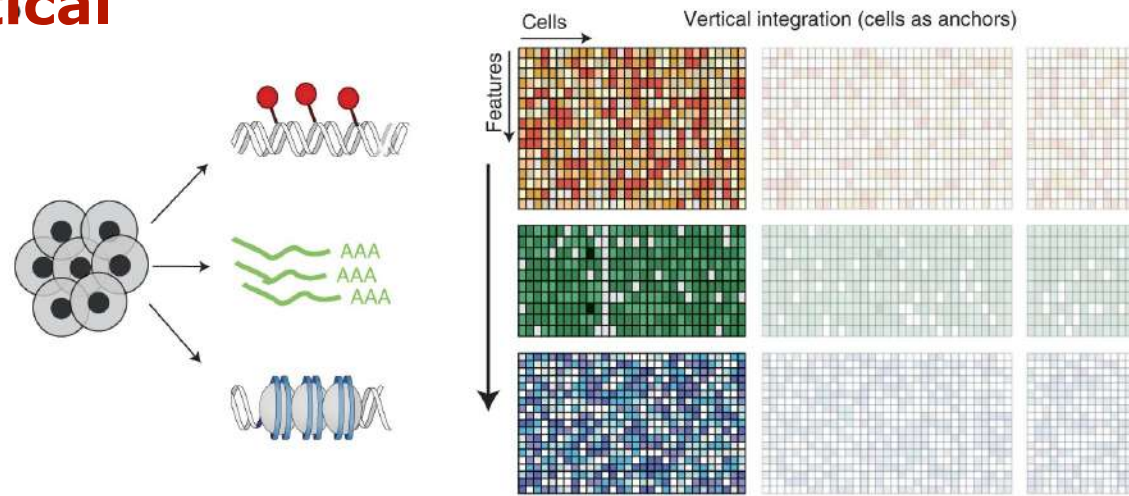
Same-section measurements: different features measured on the exact same cells. Observations are already matched

Uni-modal sections: same set of features is measured across sections. Spots are not paired across slices, so a cell-cell or spot-spot matching must be established before integration

Multi-modal sections: different features on different sections. Sections must be spatially aligned before any joint analysis can be performed

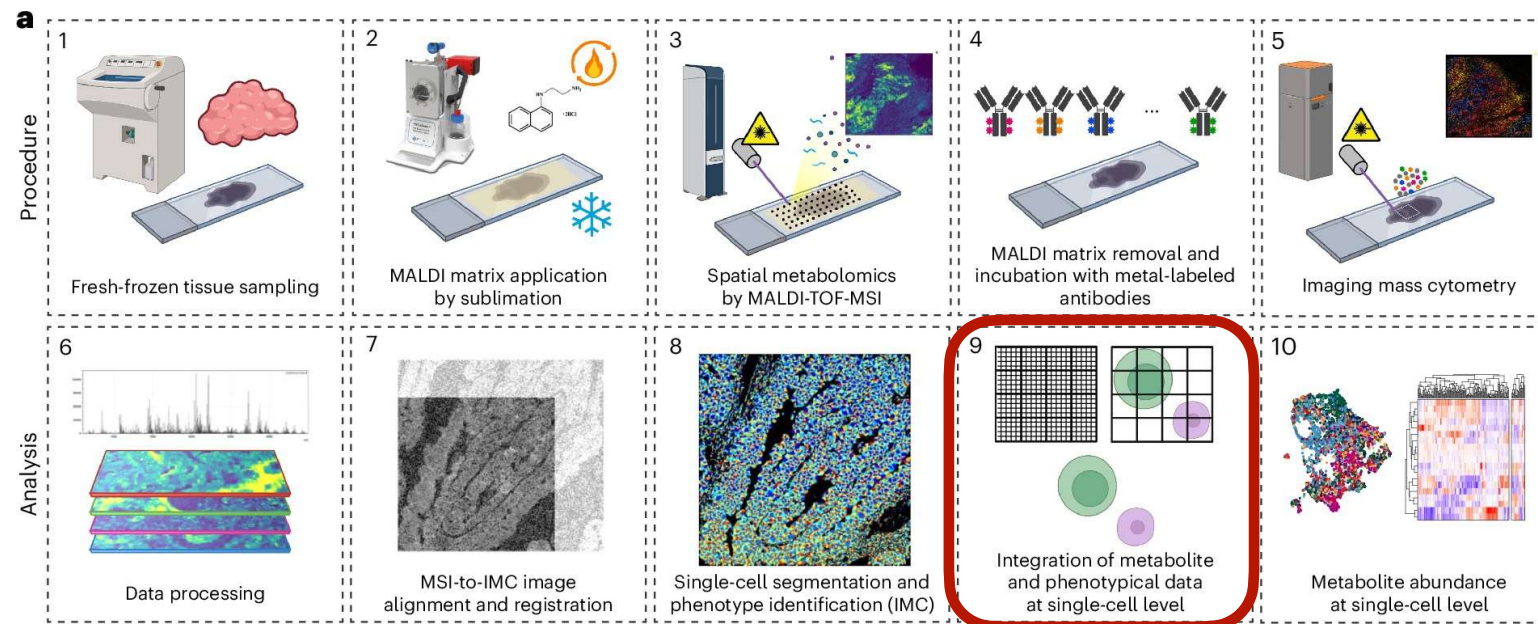
Vertical integration requires minimal alignment

1 Vertical



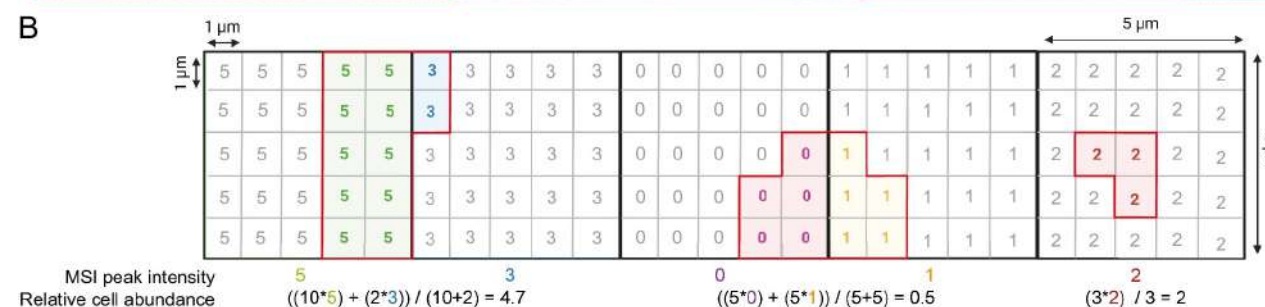
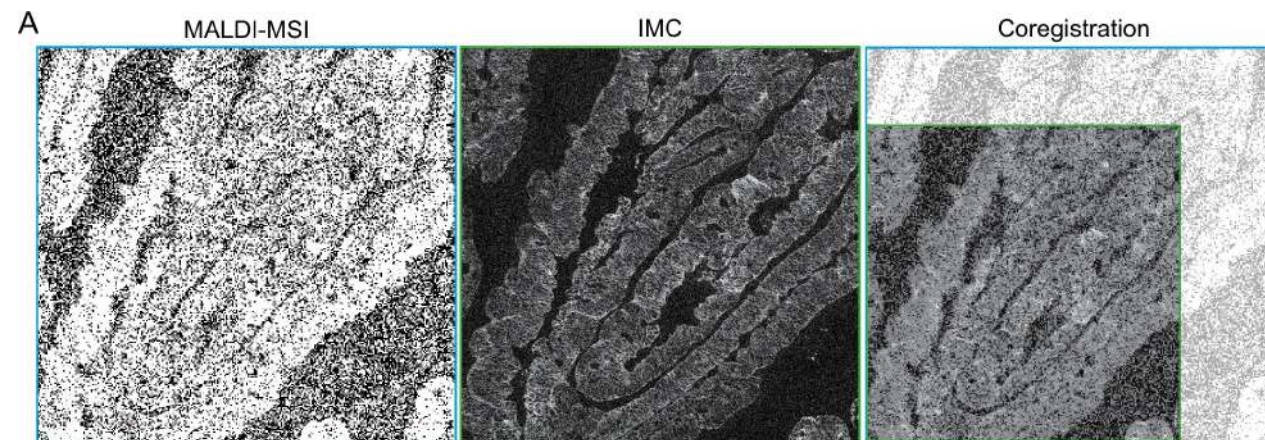
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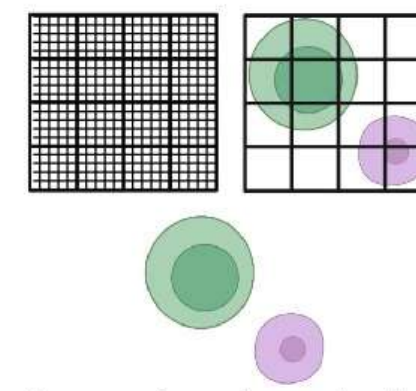
- Differences in resolution
- Differences in measured area

From Nunes-Ijsselstein et al. 2024 (Nature Methods)



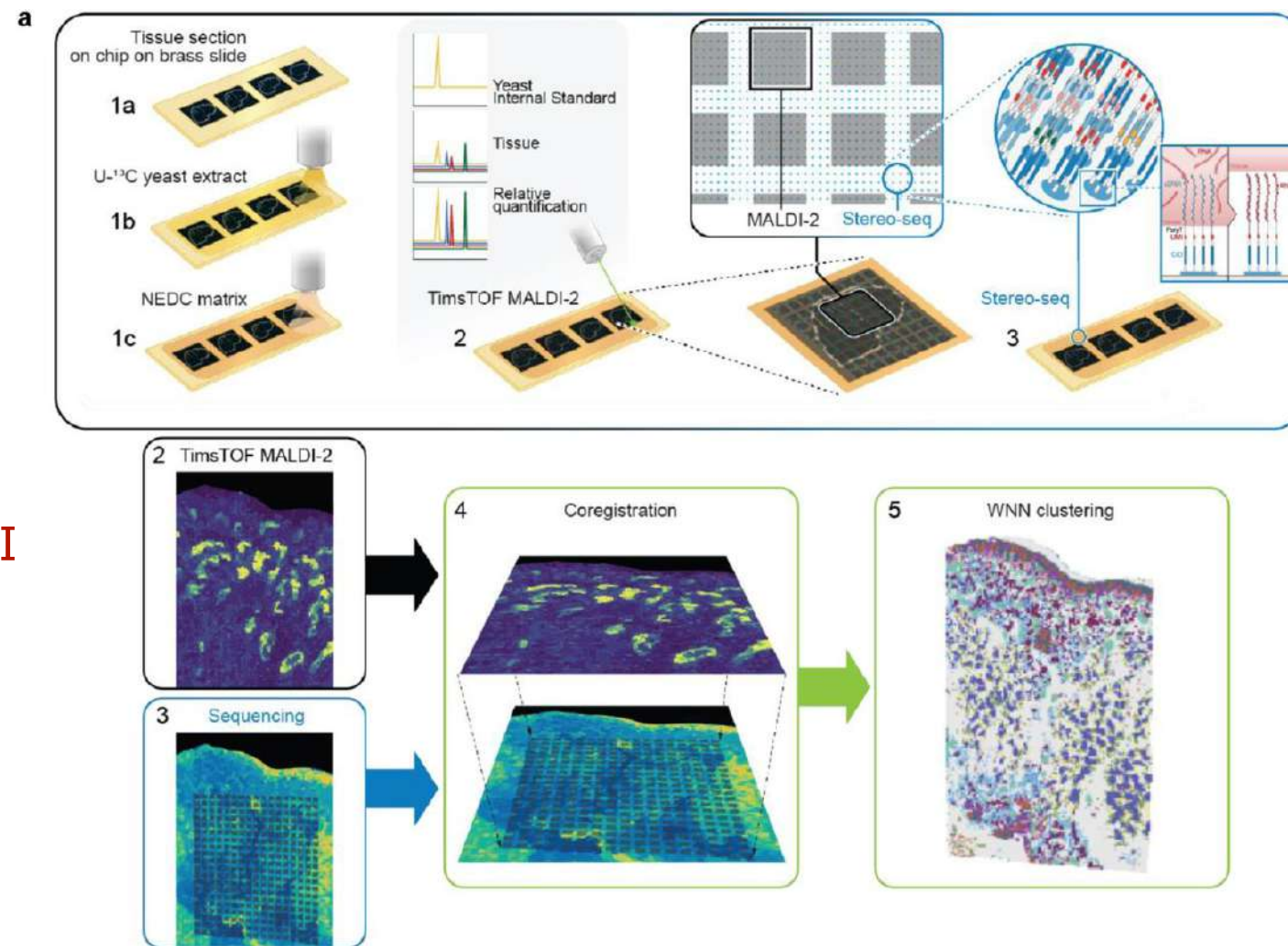
1 Manual co-registration

2 Fix difference in resolution



Vertical integration requires minimal alignment

Same section measurement of spatial transcriptomics (Stereo-seq) and spatial metabolomics (MALDI-MSI)



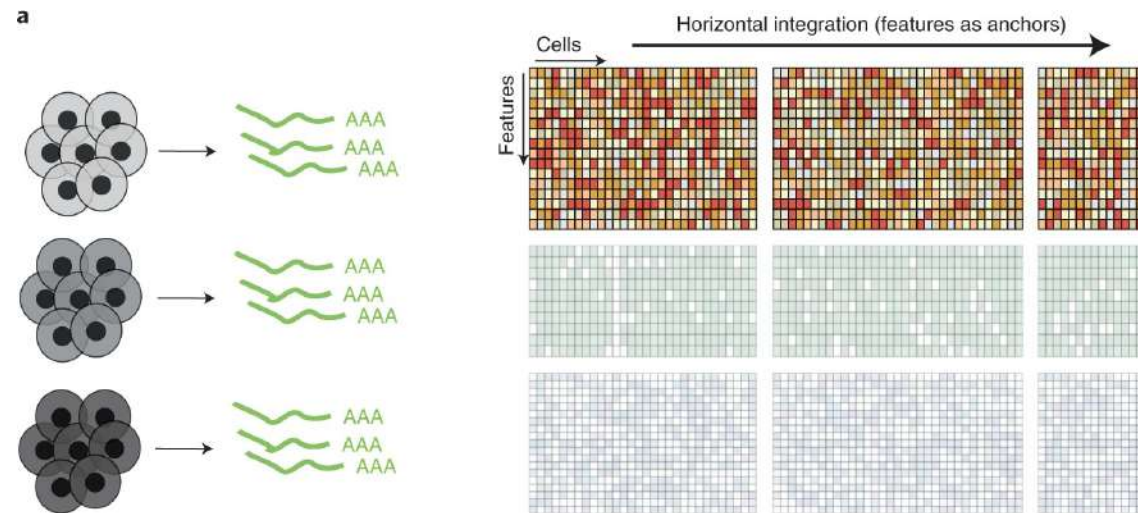
Modality 1: MALDI

Modality 2: ST

Differences in measured area: transcriptomics profiles have been measured on the entire tissue, metabolomics ones only on a portion -> needed rigid image-co registration (translation and scaling), followed by k-nearest neighbour matching of spots.

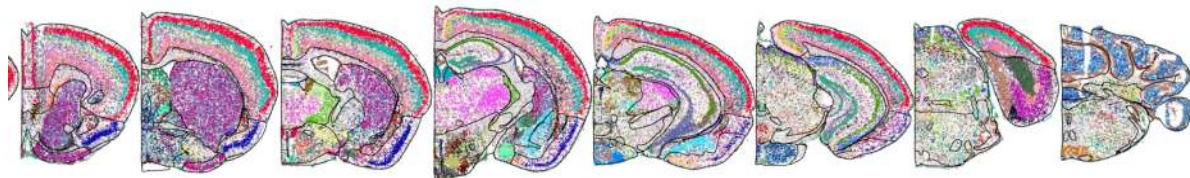
Three integration cases

2 Horizontal

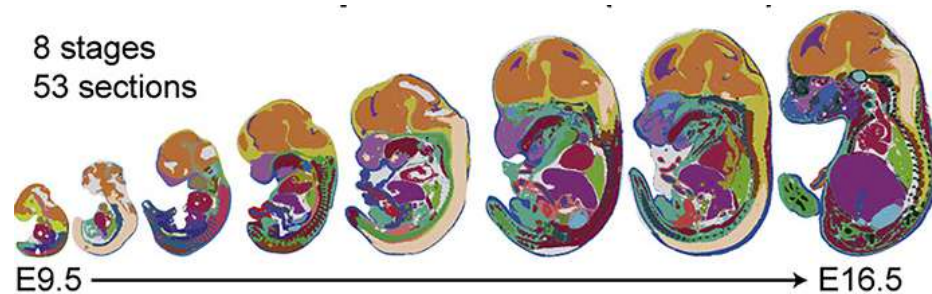


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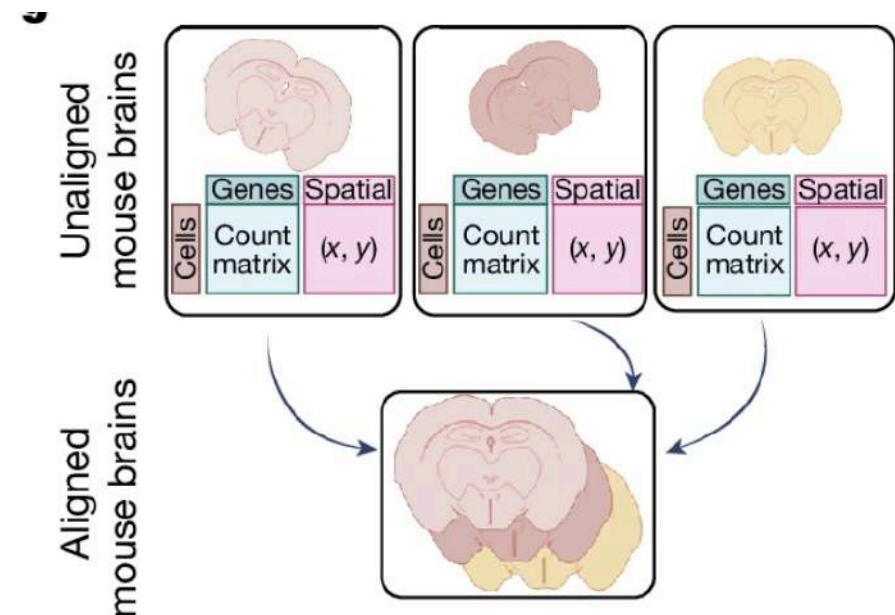
e.g. Spatial Transcriptomics



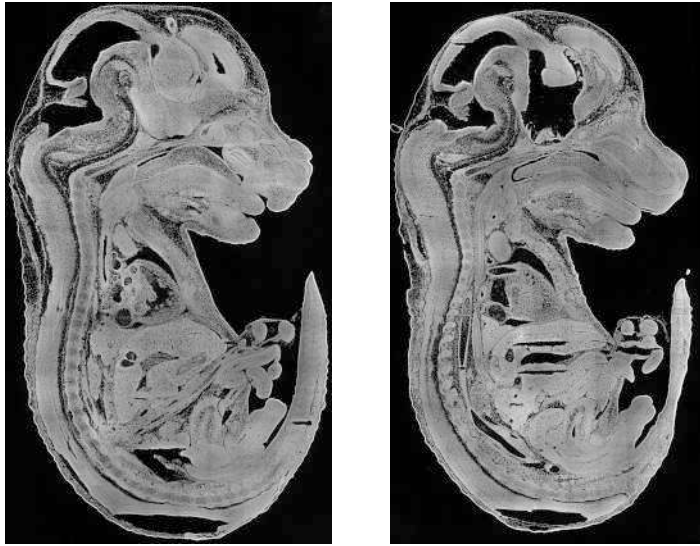
From Zhang et al. 2023 (Nature)



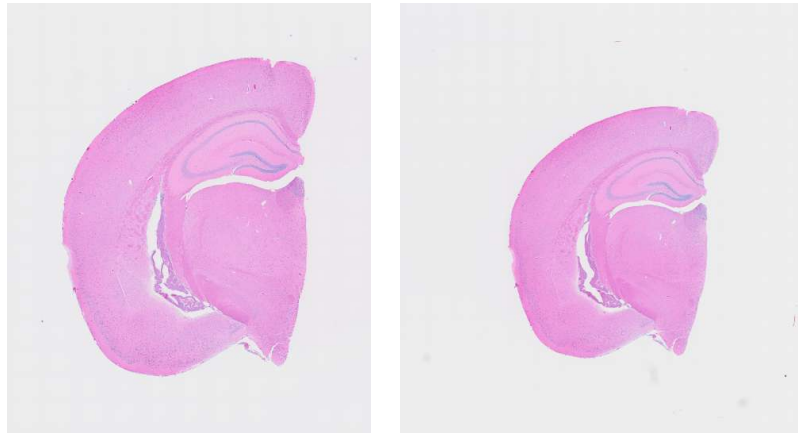
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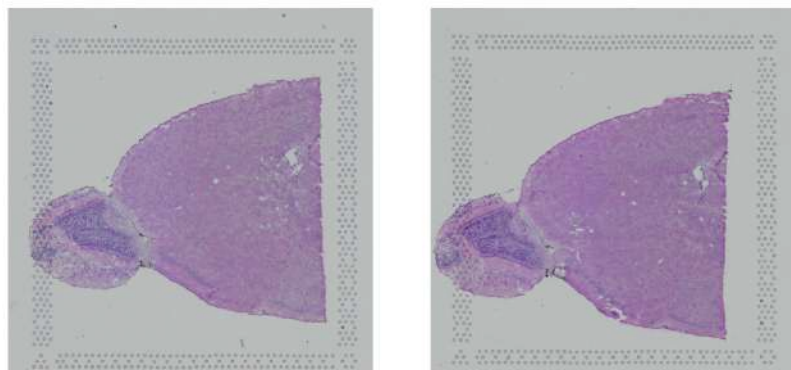
Uni-modal consecutive measurement datasets



Stereo-seq Mouse Embryo (MOSTA)

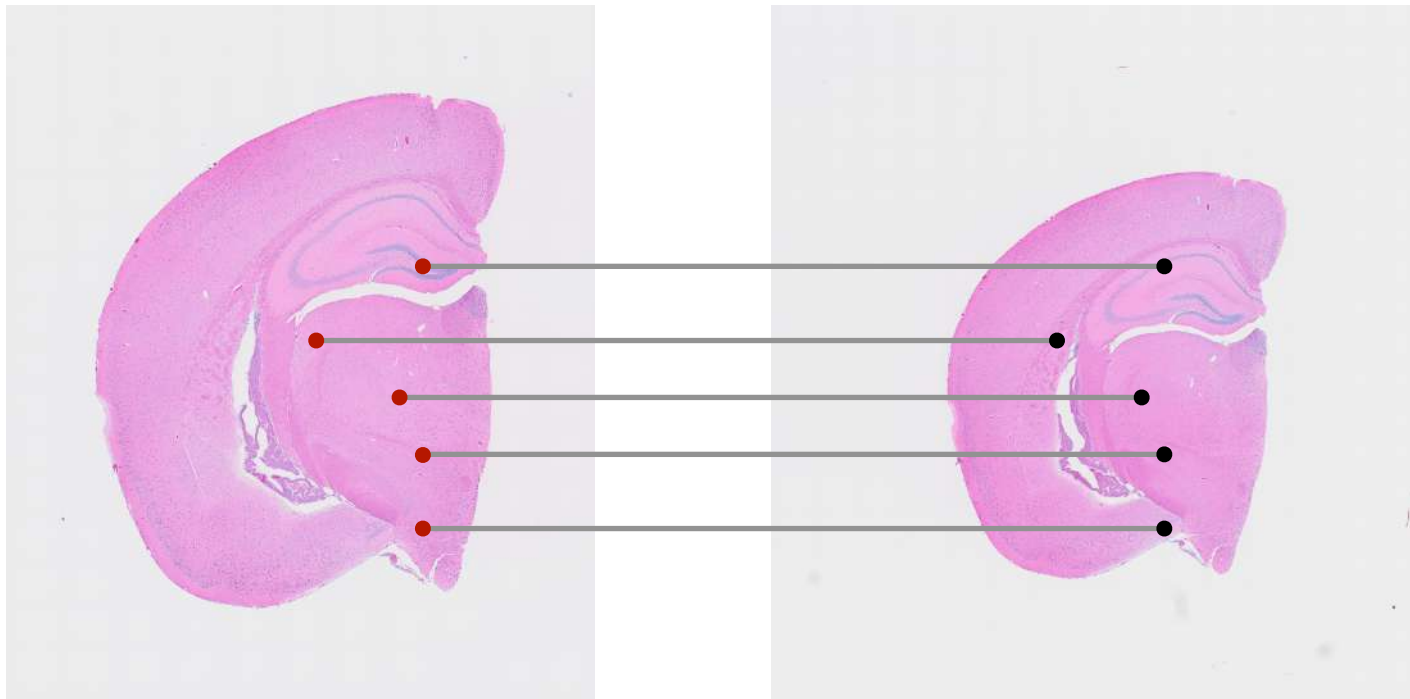


10X Visium Mouse Coronal sections



10X Visium Mouse Sagittal Anterior sections

Methods for registration of consecutive uni-modal sections



Find a measure of similarity between cells belonging to the two sections

Measure of similarity = similarity of molecular profiles

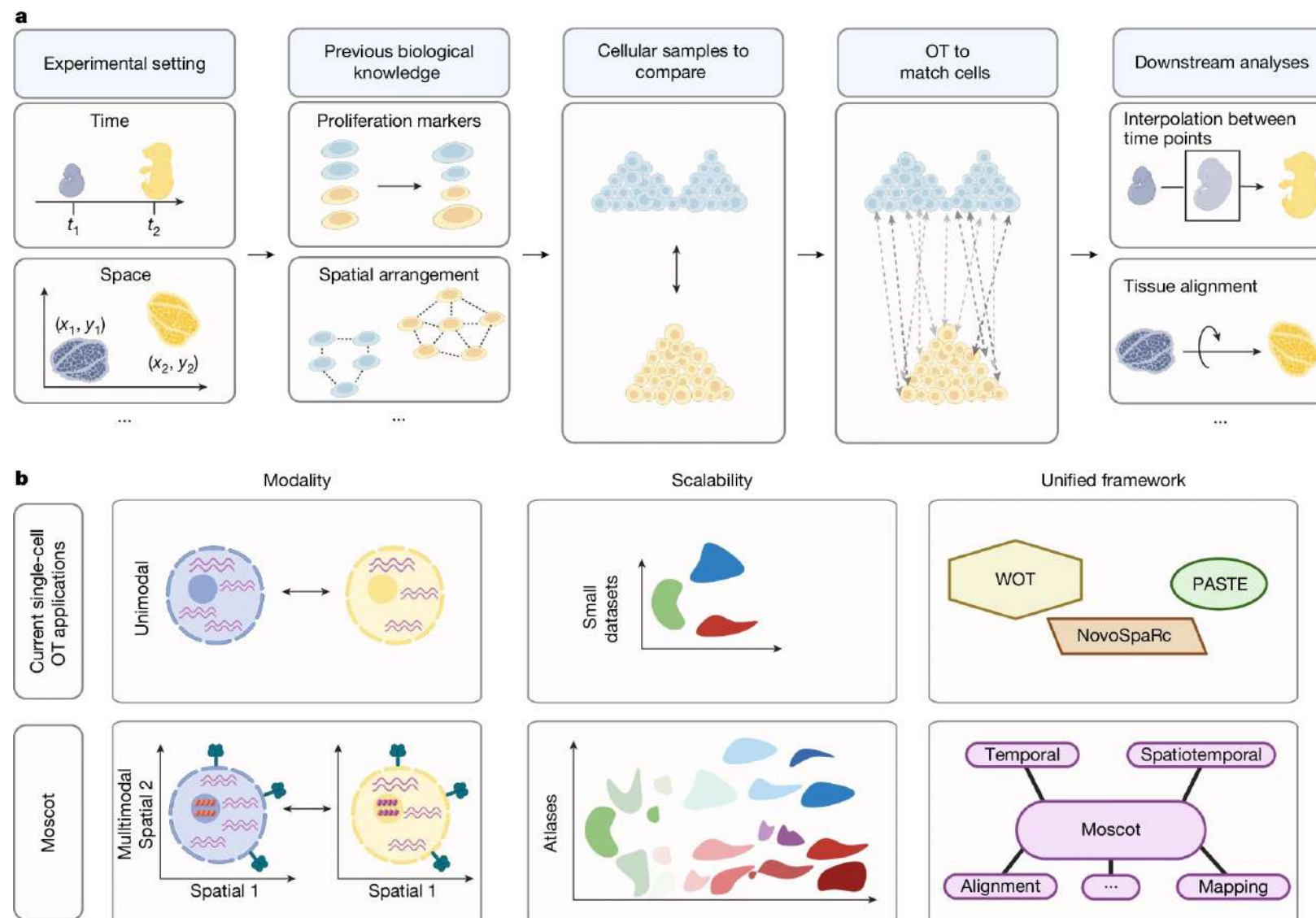


Assign best-matching cells to each other

Methods for registration of consecutive uni-modal sections

MOSCOT Multi-Omics Single-Cell Optimal Transport. Klein et al. 2025 (Nat)

- Unified OT framework with three formulations: Wasserstein (W, expression only), Gromov-Wasserstein (GW, spatial only) and Fused GW (FGW, expression + space)

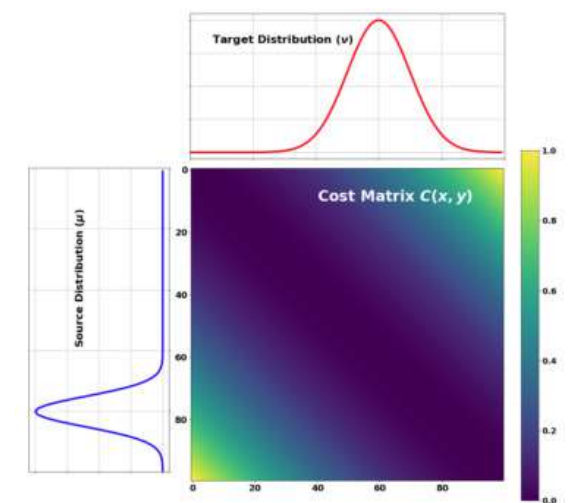
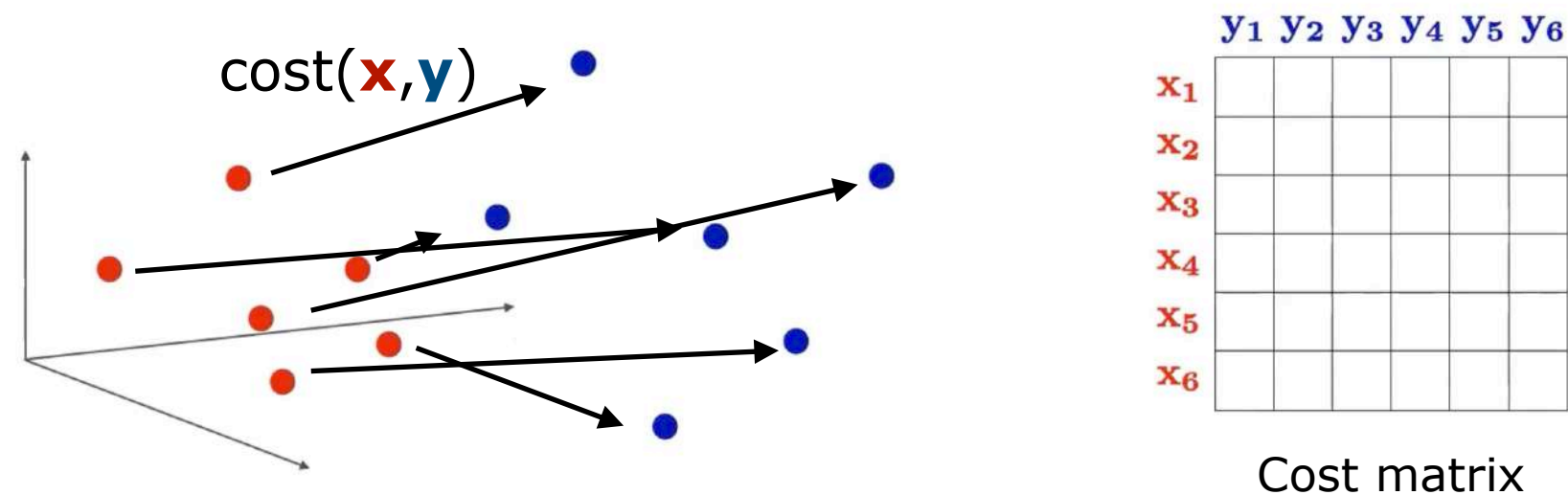


Methods for registration of consecutive uni-modal sections

MOSCOT Multi-Omics Single-Cell Optimal Transport. Klein et al. 2025 (Nat)

Parenthesis about Optimal Transport (OT)

Mathematical problem of moving one distribution of mass to another with the least possible effort

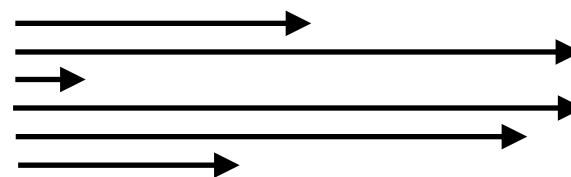


(a) An example of the cost matrix (ground metric) that can be chosen to weight the "cost of mass transport" from one probability distribution into another.

Find the optimal match by minimising the cost

$$\min_{\sigma \in \mathcal{S}_n} \sum_{i=1}^n c(x_i, y_{\sigma_i})$$

= sum of distances



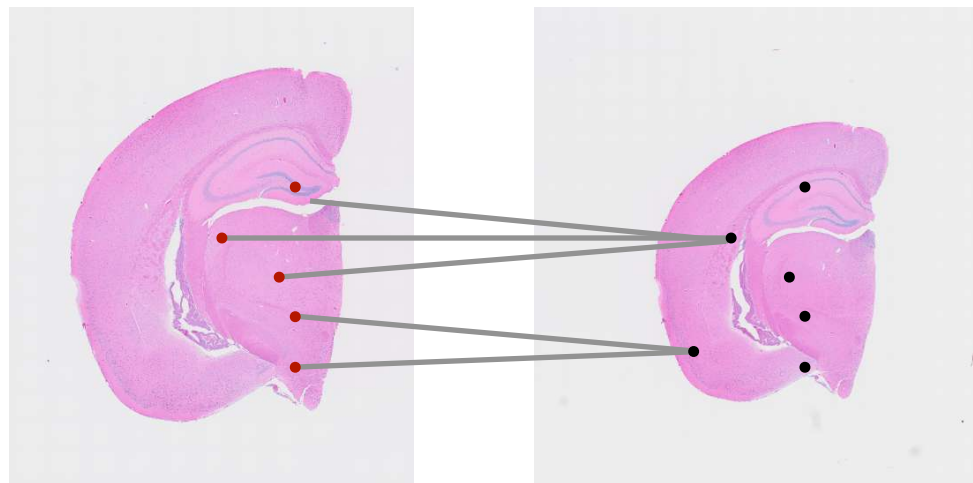
+ physical cost matrix

Methods for registration of consecutive uni-modal sections

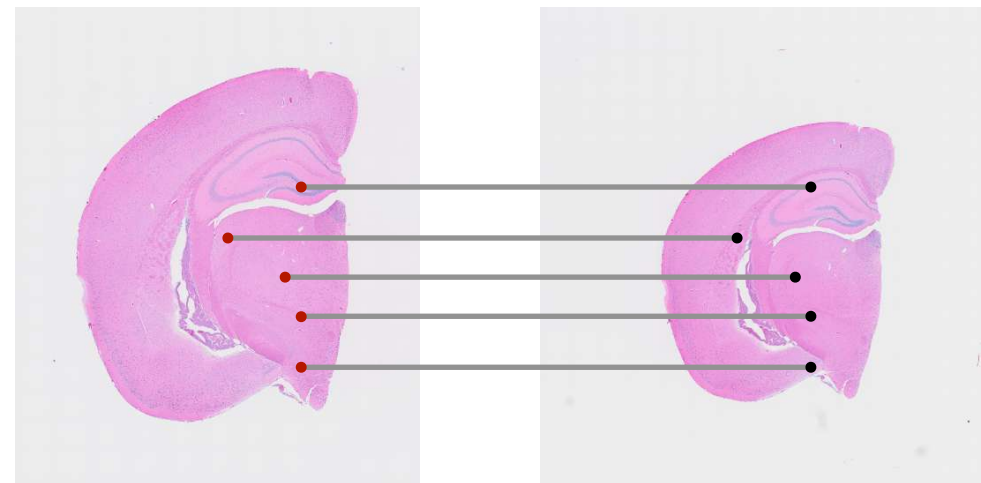
MOSCOT Multi-Omics Single-Cell Optimal Transport. Klein et al. 2025 (Nat)

Parenthesis about Optimal Transport (OT)

Regularisation to make it less computationally expensive and to penalise low-entropy plans (= transport plans where 1 “donor” cell in dataset1 is the best donor for most cells in datasets2)



Low entropy plan

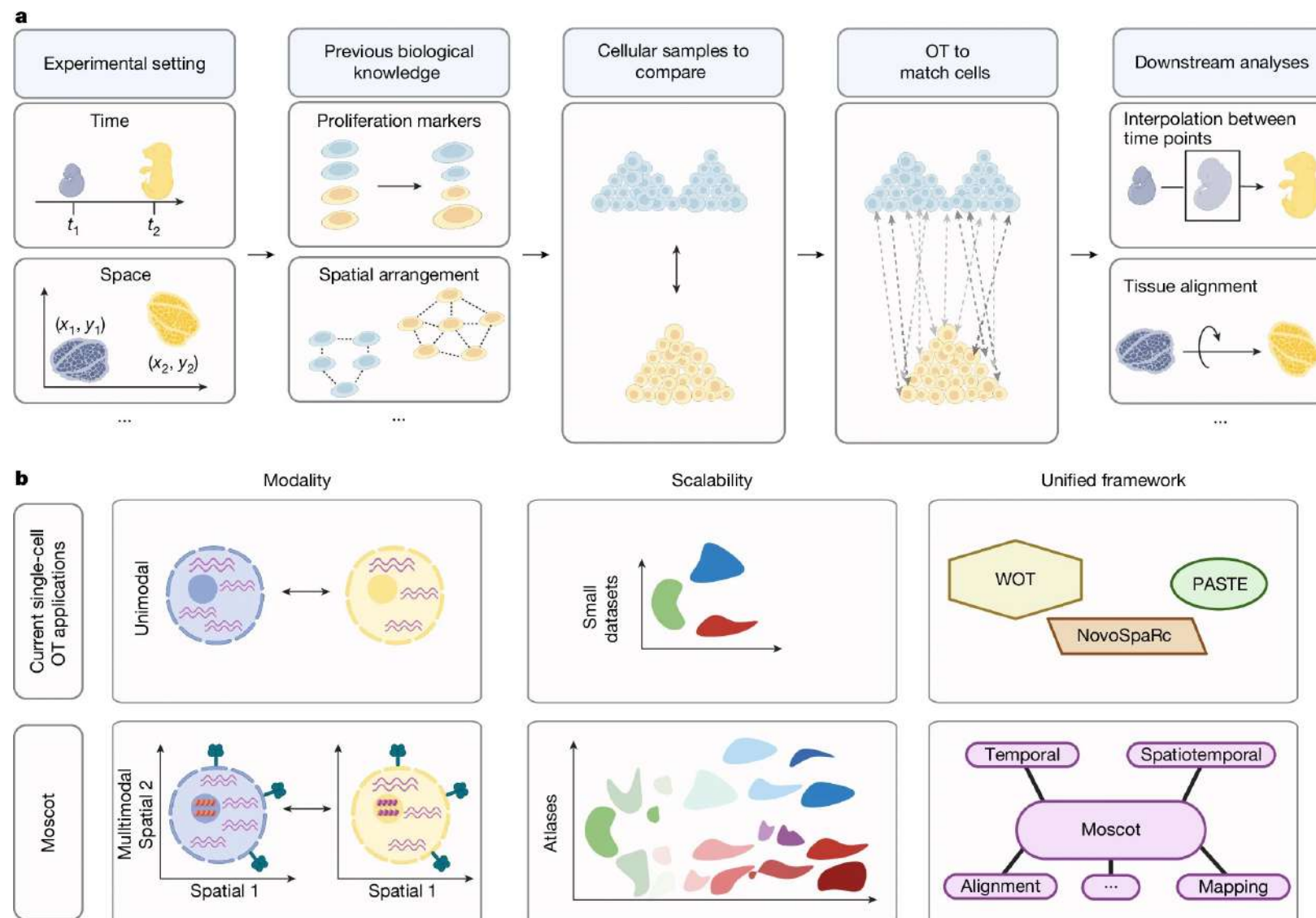


High entropy plan

Methods for registration of consecutive uni-modal sections

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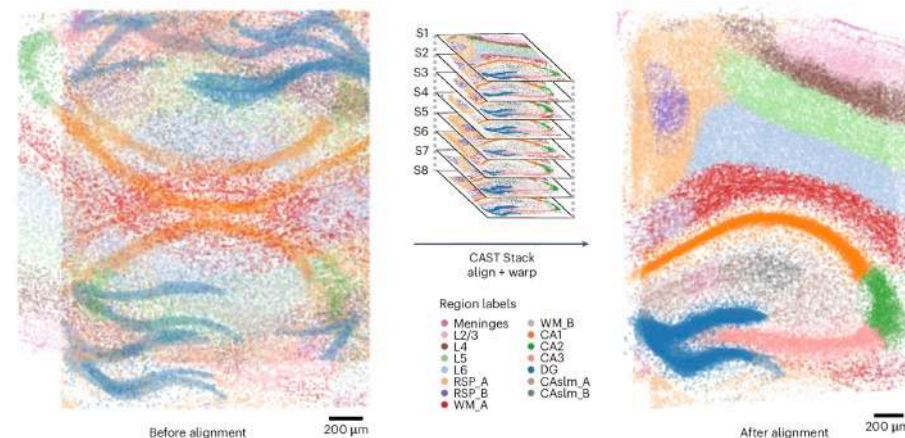
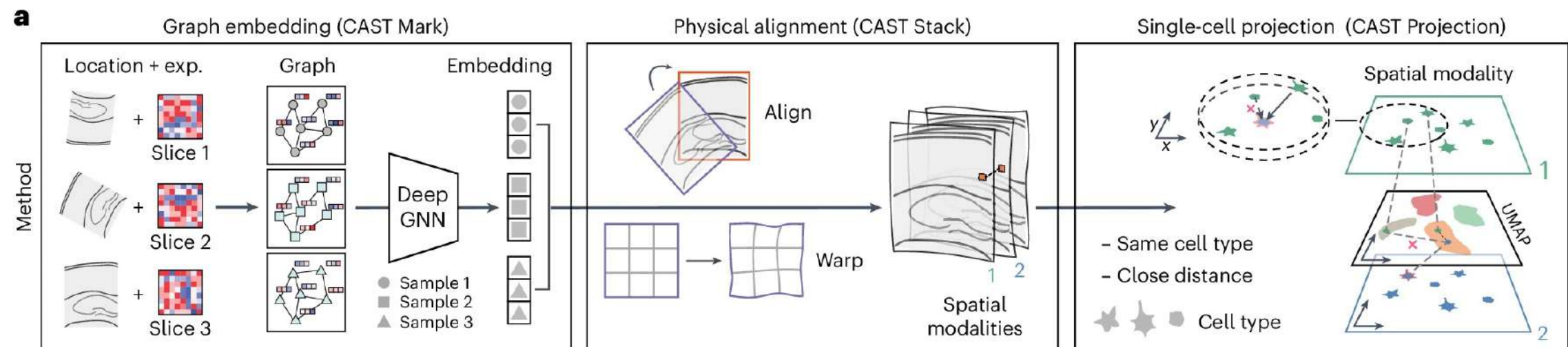
- Unified OT framework with three formulations: Wasserstein (W, expression only), Gromov-Wasserstein (GW, spatial only) and Fused GW (FGW, expression + space)



Methods for registration of consecutive uni-modal sections

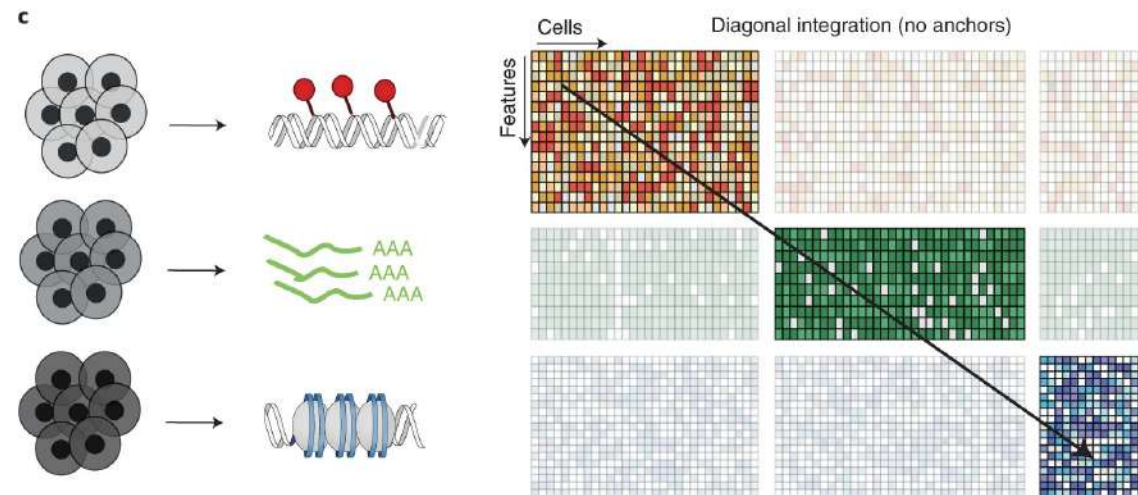
CAST Cross-sample Alignment of Spatial Transcriptomics. Tang et al. 2024 (Nat Met)

- Three-module pipeline:
 - CAST Mark (self-supervised GNN embedding combining expression + spatial graph),
 - CAST Stack (cross-sample alignment by intrinsic molecular similarity),
 - CAST Projection (spatially resolved multi-omic profiles)



Three integration cases

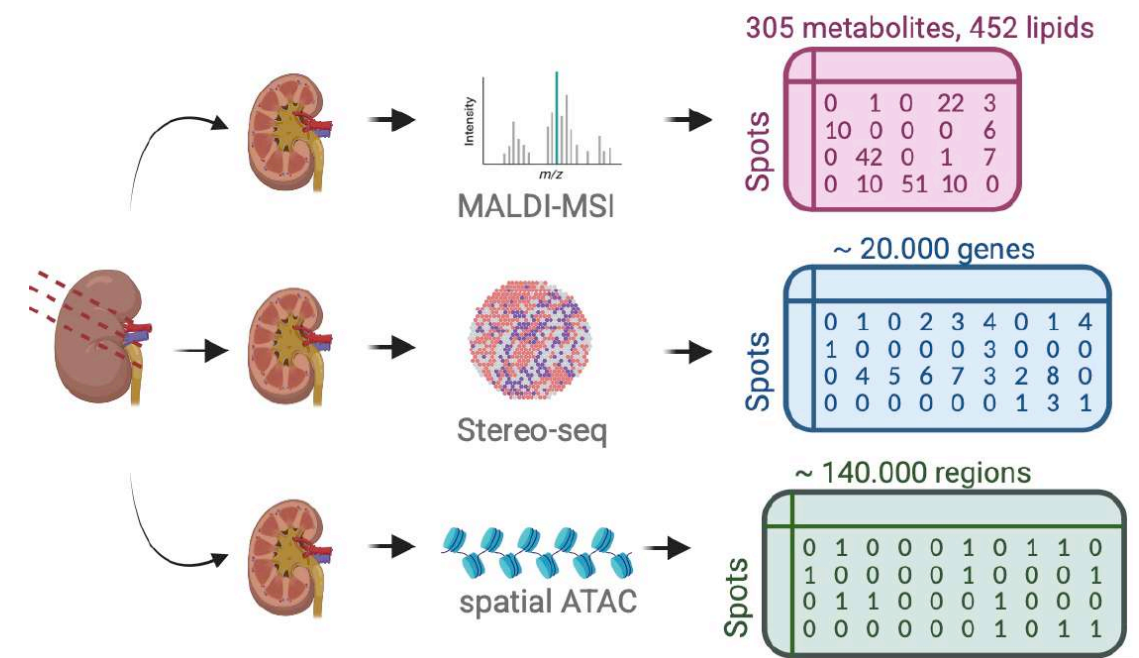
3 Diagonal



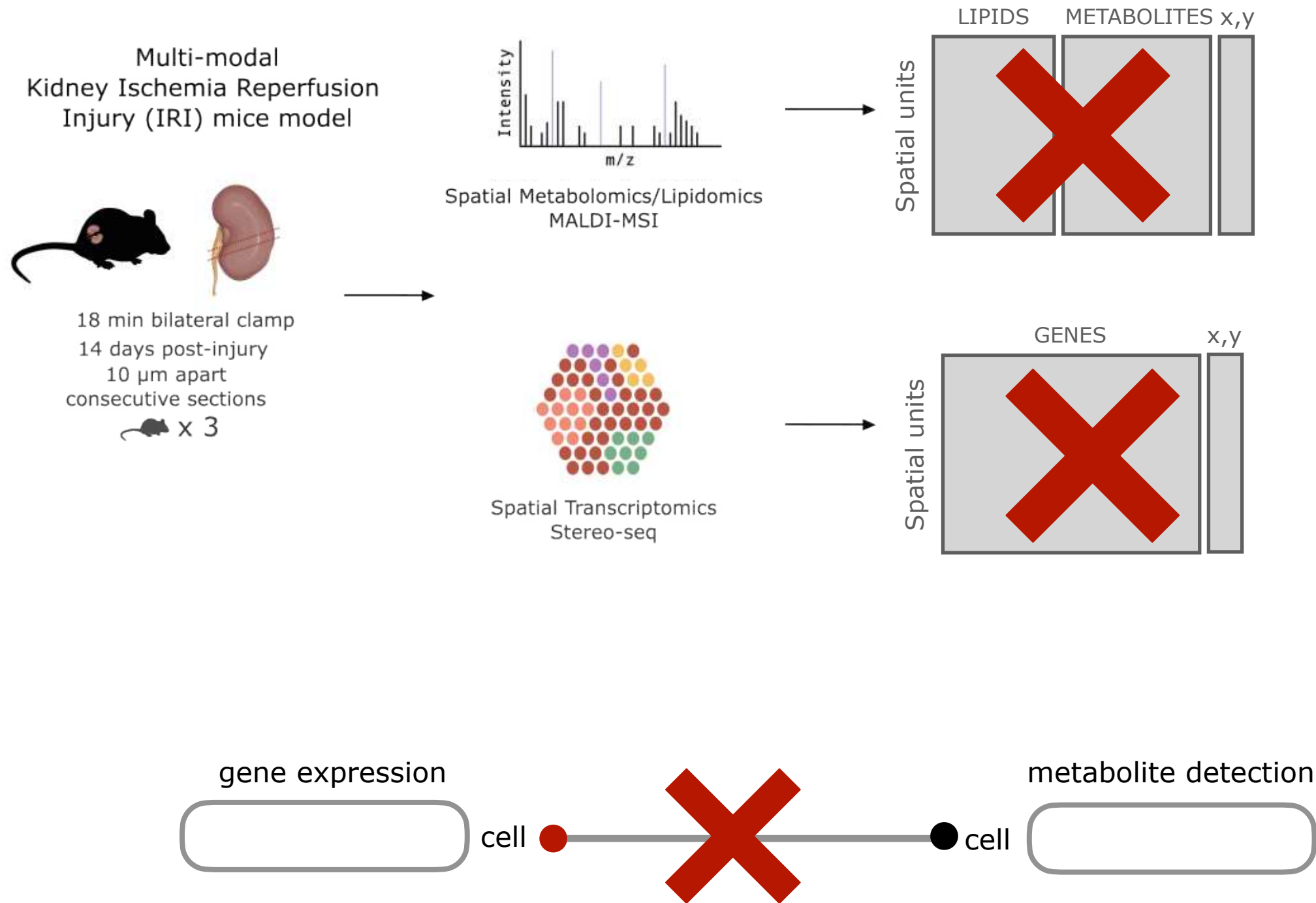
Multi-modal sections: different features on different sections. Sections must be spatially aligned before any joint analysis can be performed

Different molecular features measured
(= multi-modalities / multi-omics)
e.g. spatial transcriptomics + spatial metabolomics

Different molecular layers = different views of the same tissue



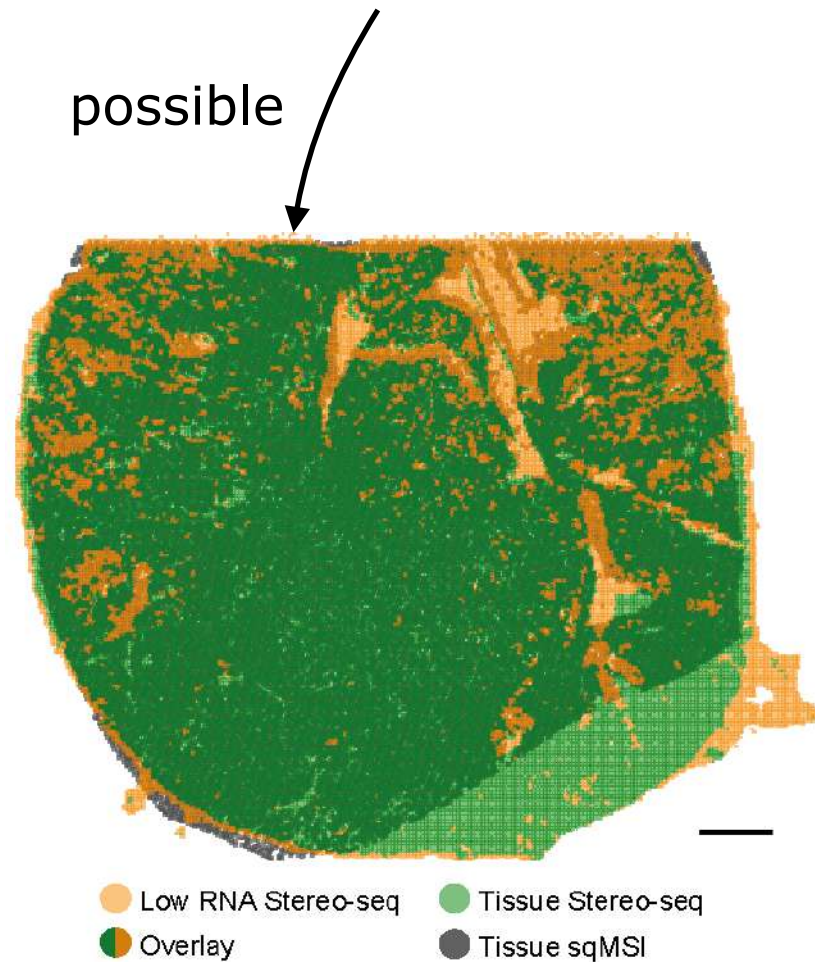
Multi-modal alignment cannot use molecular features



Multi-modal alignment cannot use molecular features

Solution: use images/cell coordinates

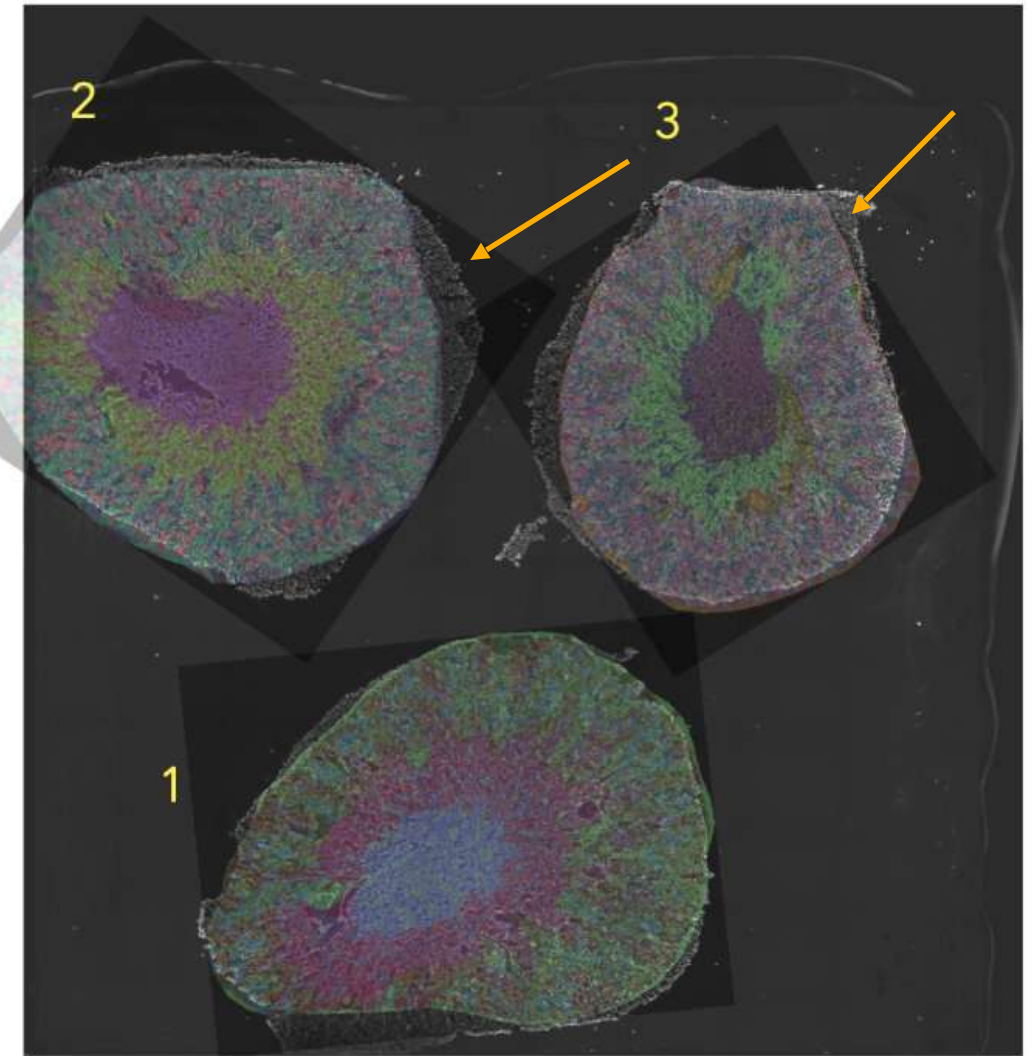
- Rigid transformations (rotation, translation, scaling). Potentially fast but limited to non-distorted sections



Green = Spatial Transcriptomics (Stereo-seq)
Orange = Spatial Metabolomics/Lipidomics (qMSI)

Rietjens et al. 2026 (JASN)

not recommended

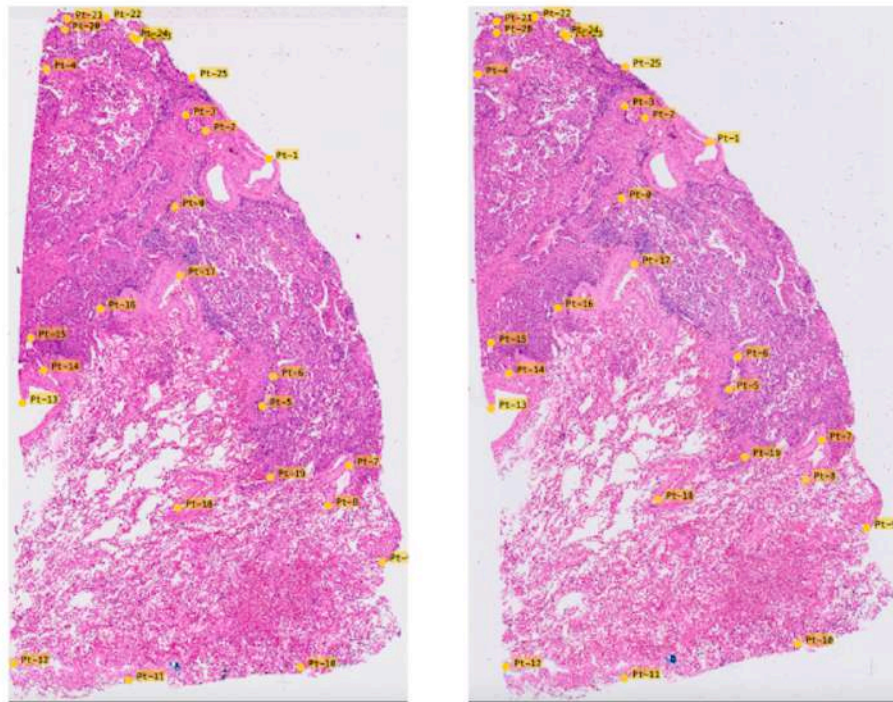


Colored = modality 1
Greyscale = modality 2

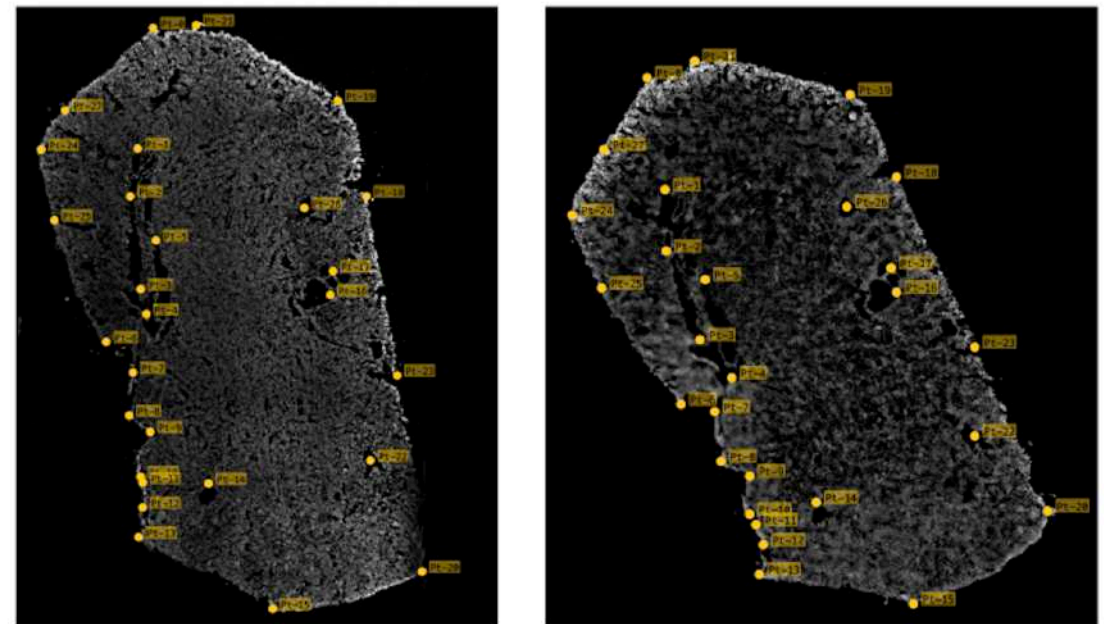
Multi-modal alignment cannot use molecular features

Solution: use images/cell coordinates

- Rigid transformations (rotation, translation, scaling). Potentially fast but limited to non-distorted sections
- Landmark-based: manually or automatically annotated anatomical landmarks guide alignment



10X Xenium on Human Lung Cancer consecutive sections



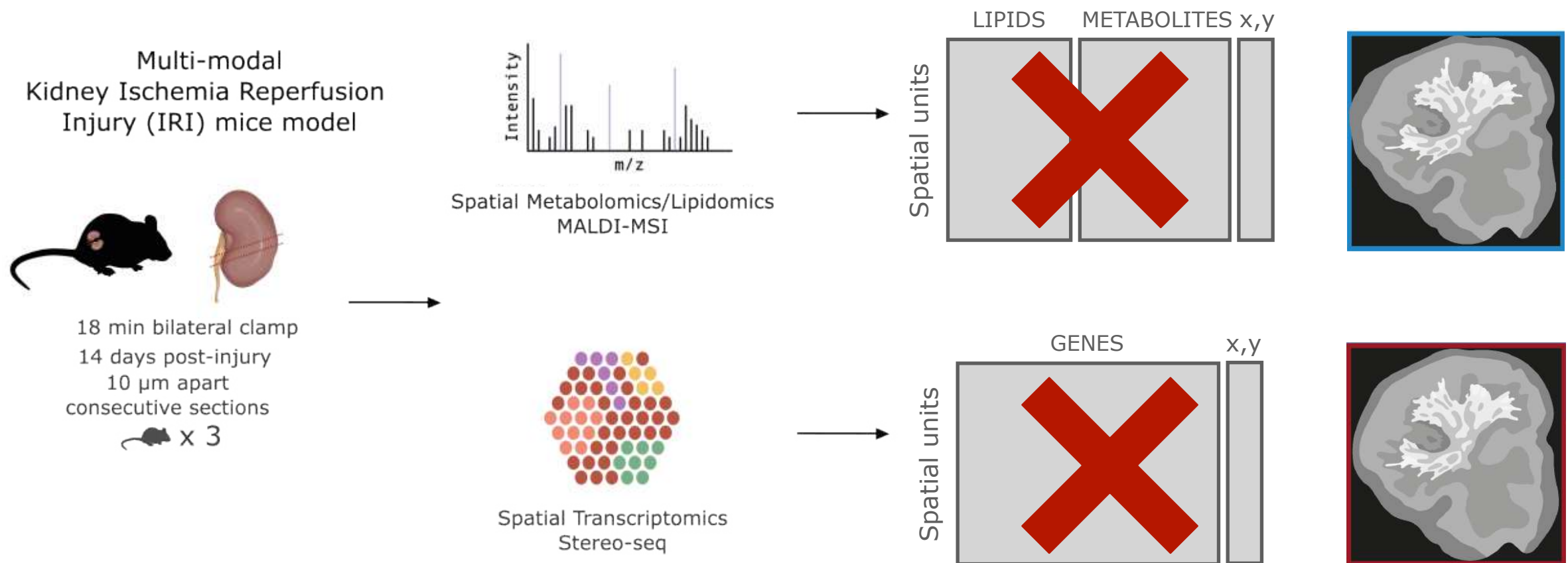
Stereo-seq and Mass spectrometry Imaging on Mouse Ischemia Reperfusion Injury Kidney

Multi-modal alignment cannot use molecular features

Solution: use images/cell coordinates

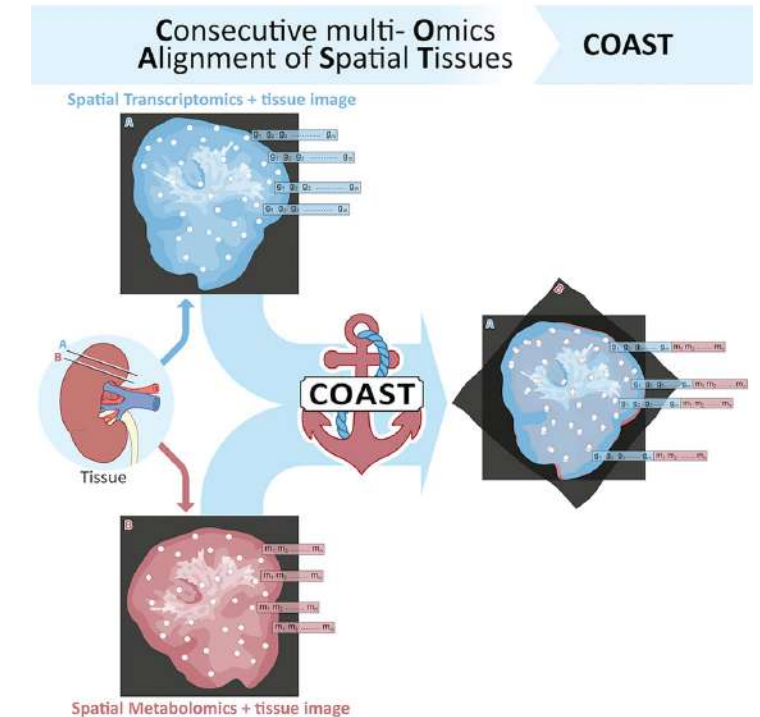
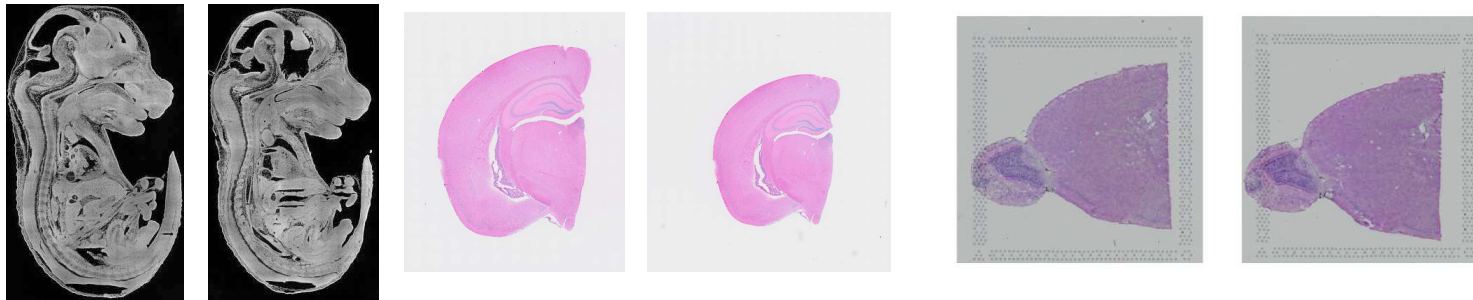
- Rigid transformations (rotation, translation, scaling). Potentially fast but limited to non-distorted sections
- Landmark-based: manually or automatically annotated anatomical landmarks guide alignment
- Transformation based on the image features of the image associated with spatial data

 = Image

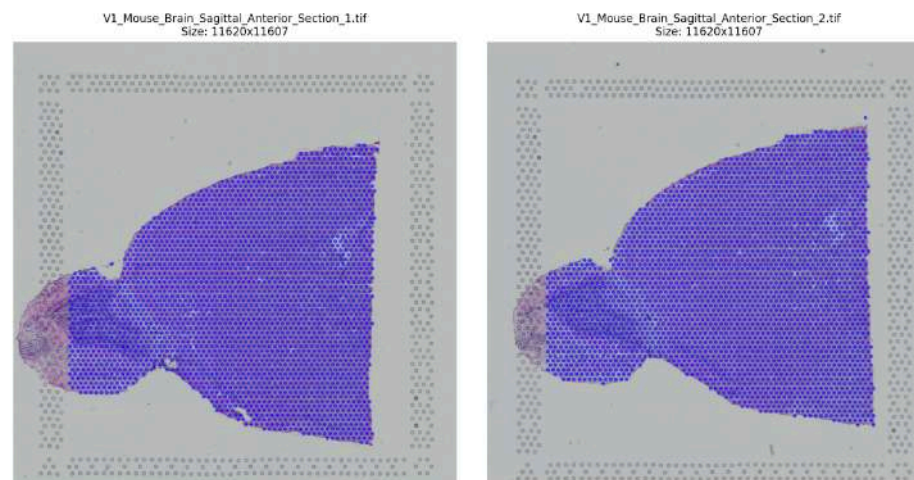


COAST uses image features

- Aligns uni- or multi-modal consecutive sections using exclusively images and coordinate data (no molecular data!)
- Requires:
 - same type of staining in the two consecutive images e.g.

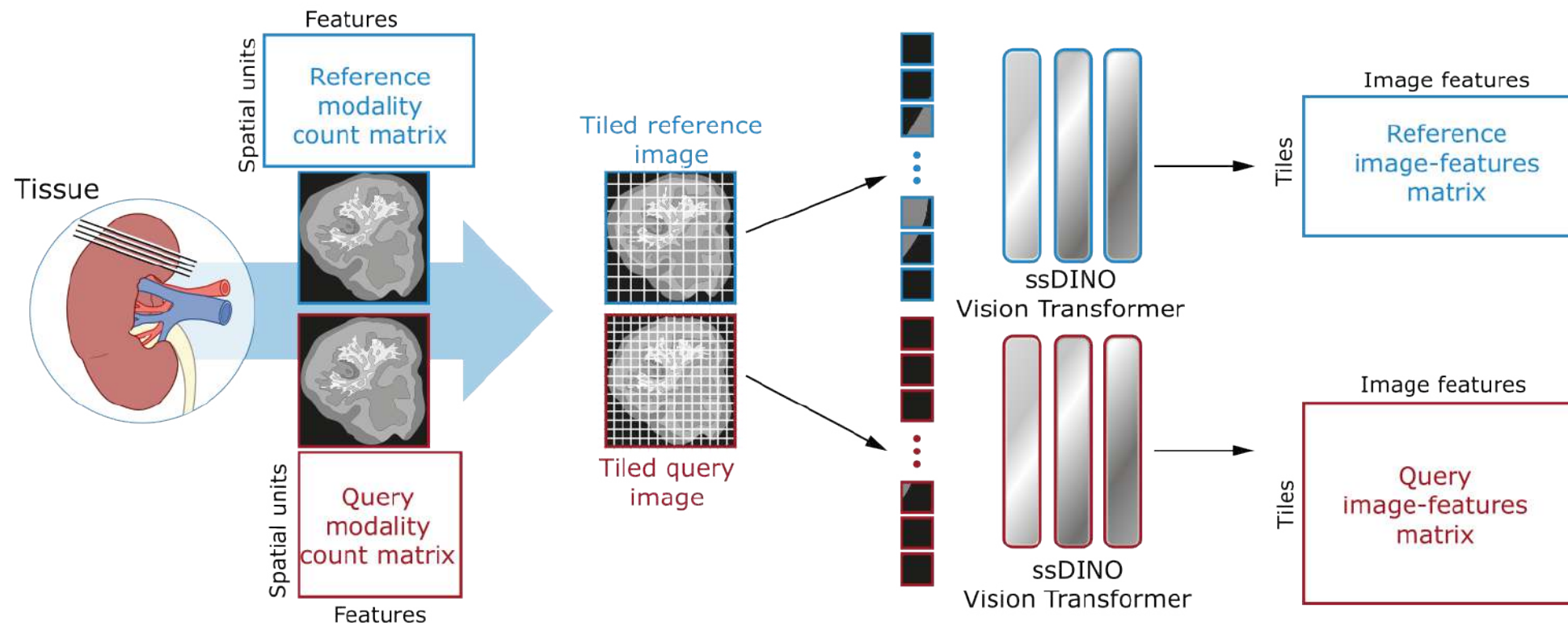


- image and x,y coordinates in the same coordinate space



- spot x,y coordinate (Visium)

COAST uses image features



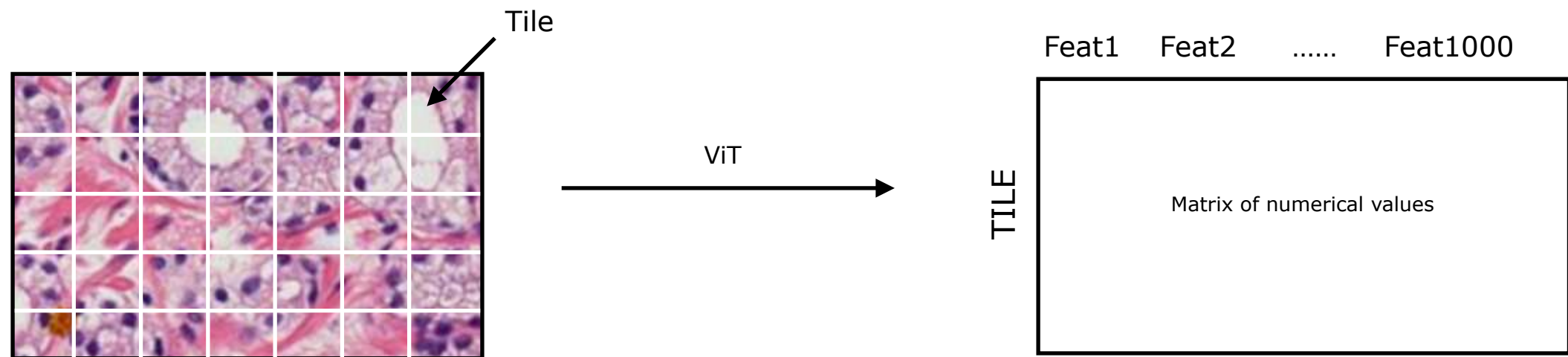
1. Tiling around each spot center
2. Extraction of image features with pre-trained Vision Transformer

Parenthesis on Vision Transformers

Generative model for images and text

The transformer encoder outputs an **embedding** = a learned high-dimensional numerical representation of the meaning of an image

"Numerical summary" that translates visual features into a language that machines can understand and compare

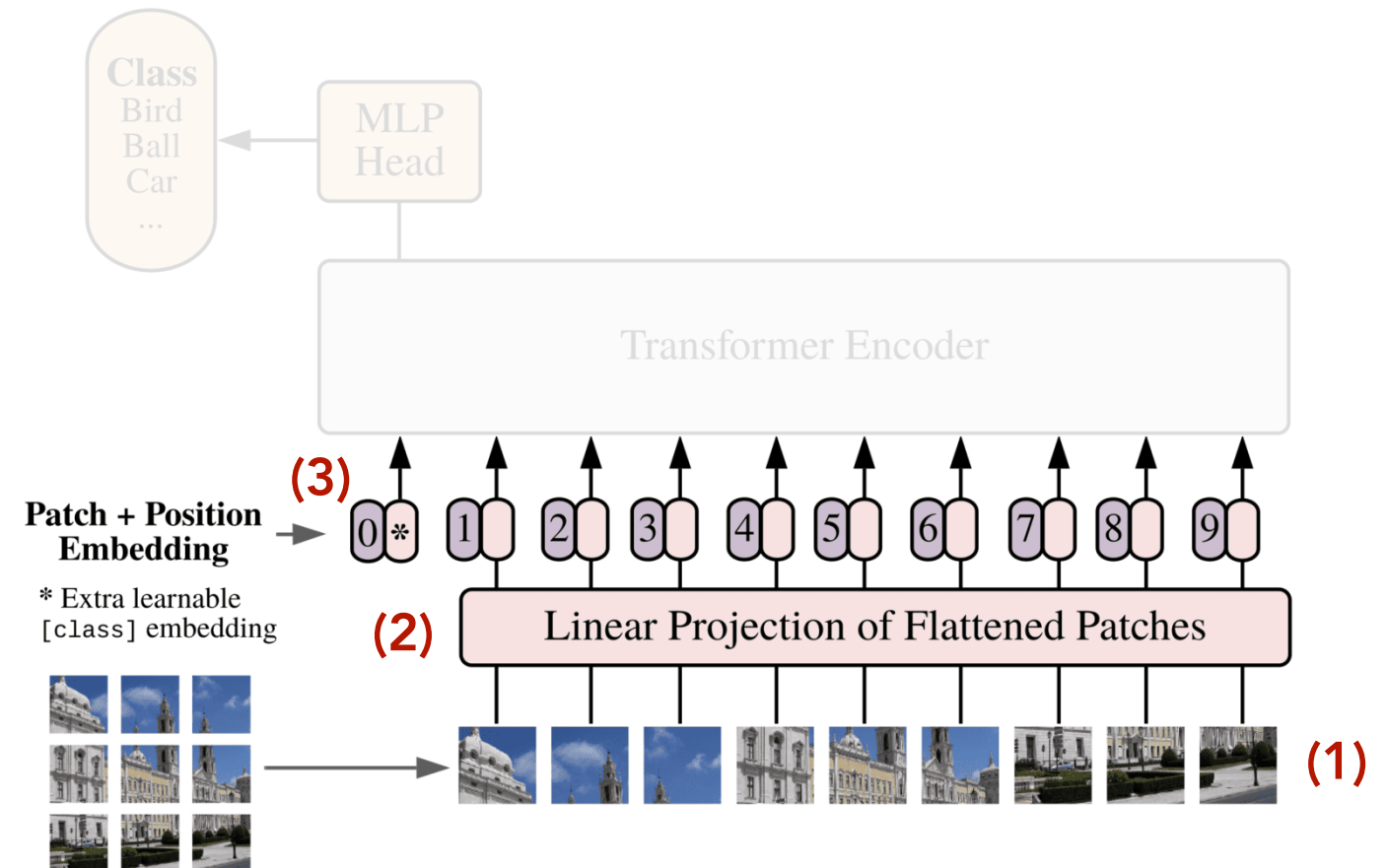


From: An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. Dosovitskiy et al. 2021

Parenthesis on Vision Transformers

Pre steps:

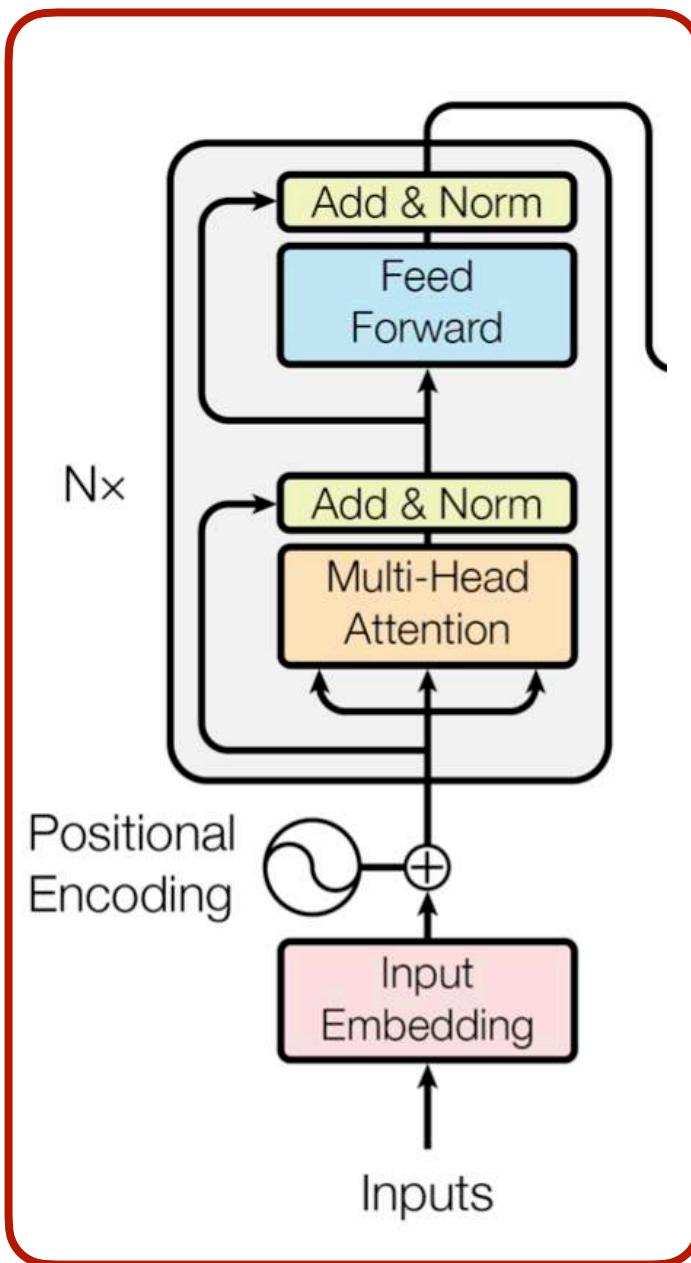
1. Tiling = the image is divided into fixed NxN tiles
2. Linear projection = each tile is mapped into a high-dimensional vector space
3. Positional encoding: 1D or 2D coordinates are added to each tile so that the model understand the image layout



From: An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. Dosovitskiy et al. 2021

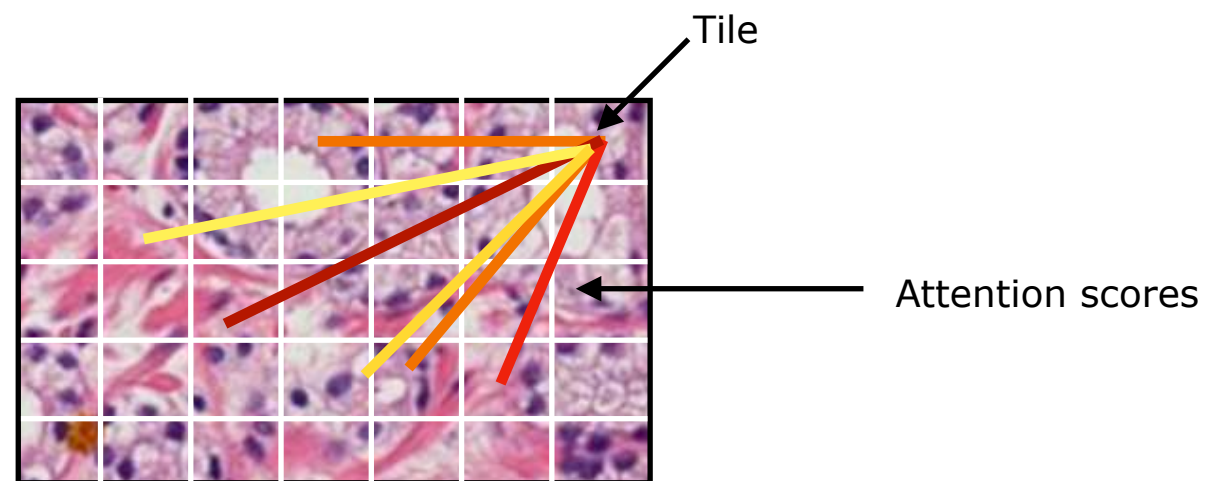
Parenthesis on Vision Transformers

Encoder block

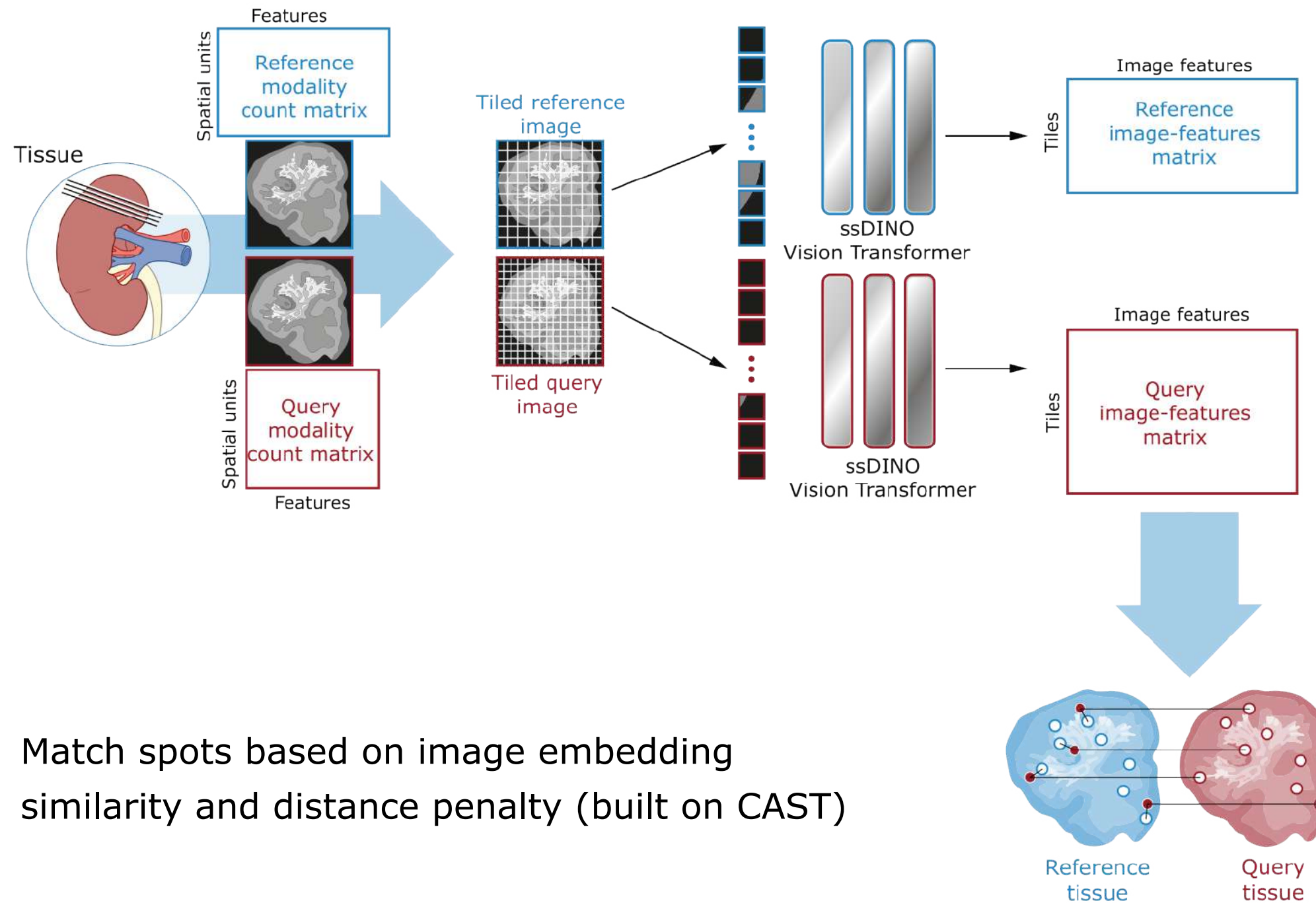


(1) Self-attention allows a transformer to decide which tiles are the most important in relations to other by calculating a score

(2) Every tile attends to each other, capturing both long- and short-range dependencies

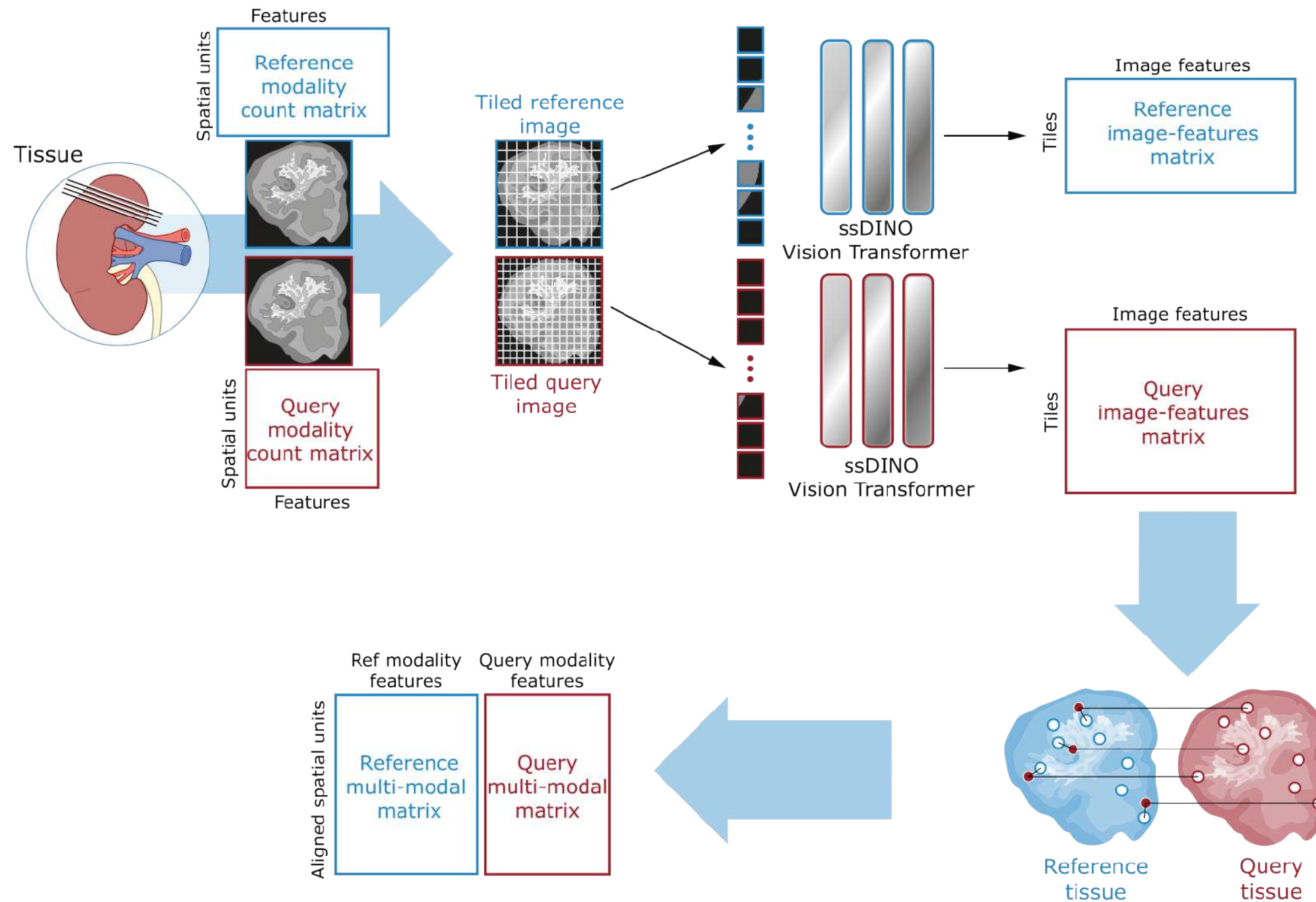


COAST uses image features



Match spots based on image embedding similarity and distance penalty (built on CAST)

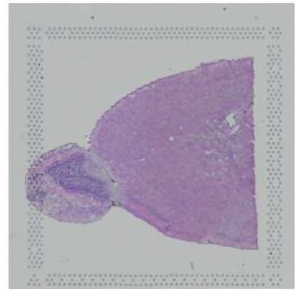
COAST uses image features



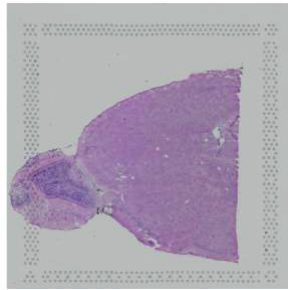
Create multi-modal dataset by assigning spots to each other with k-nearest neighbour

COAST can align uni-modal sections (as a proof of concept)

Mouse Brain Sagittal Anterior
(10X Visium)



Reference section

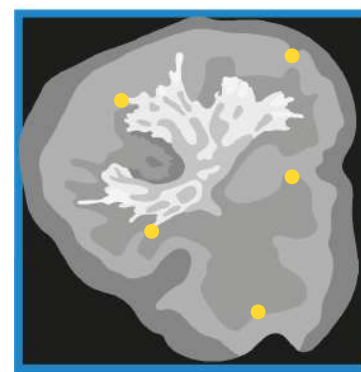


Query section

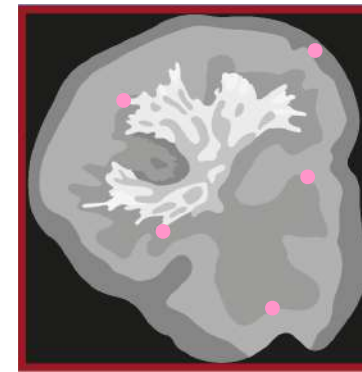


- Query section
- Reference section

Evaluation metric: landmark distance

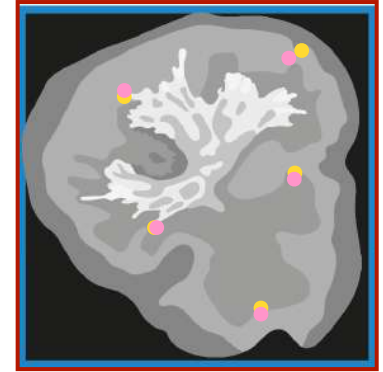


Reference section
image

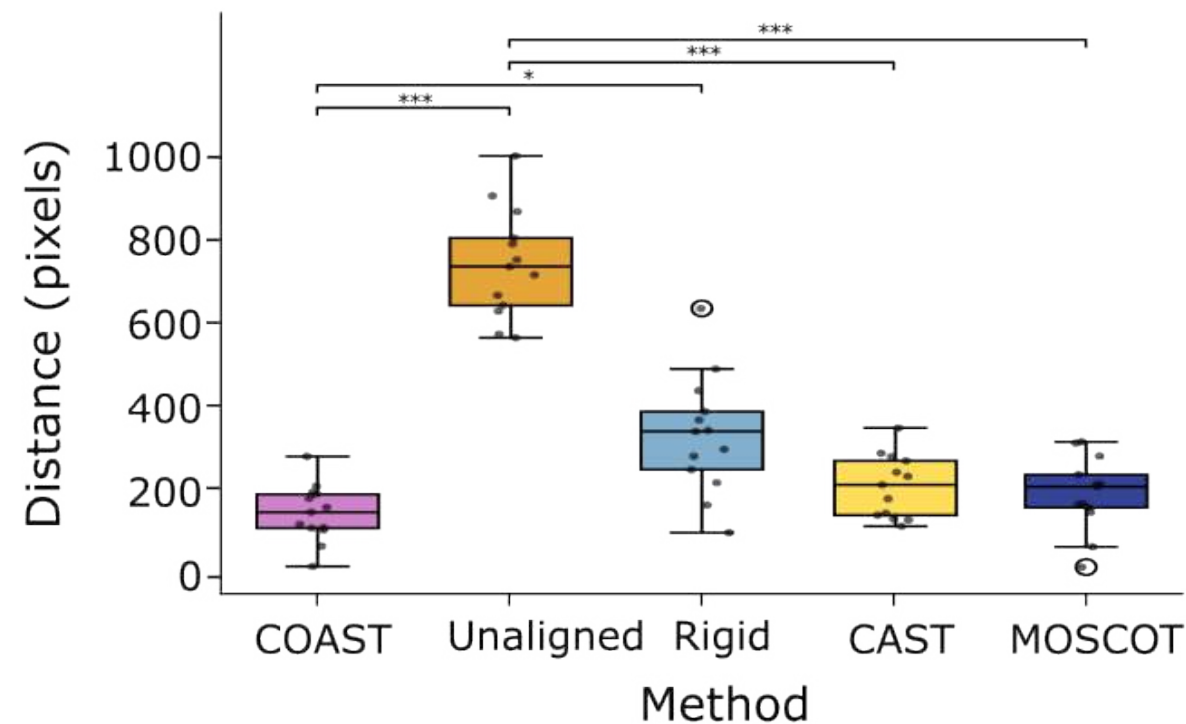


Query section
image

COAST



Query section in reference
coordinate space



COAST can align uni-modal sections (as a proof of concept)

MOSTA Mouse Embryo dataset

Spatial Transcriptomics with Stereo-seq (single-cell resolution)



E16.5_E2S6
query



E16.5_E2S7
reference

COAST

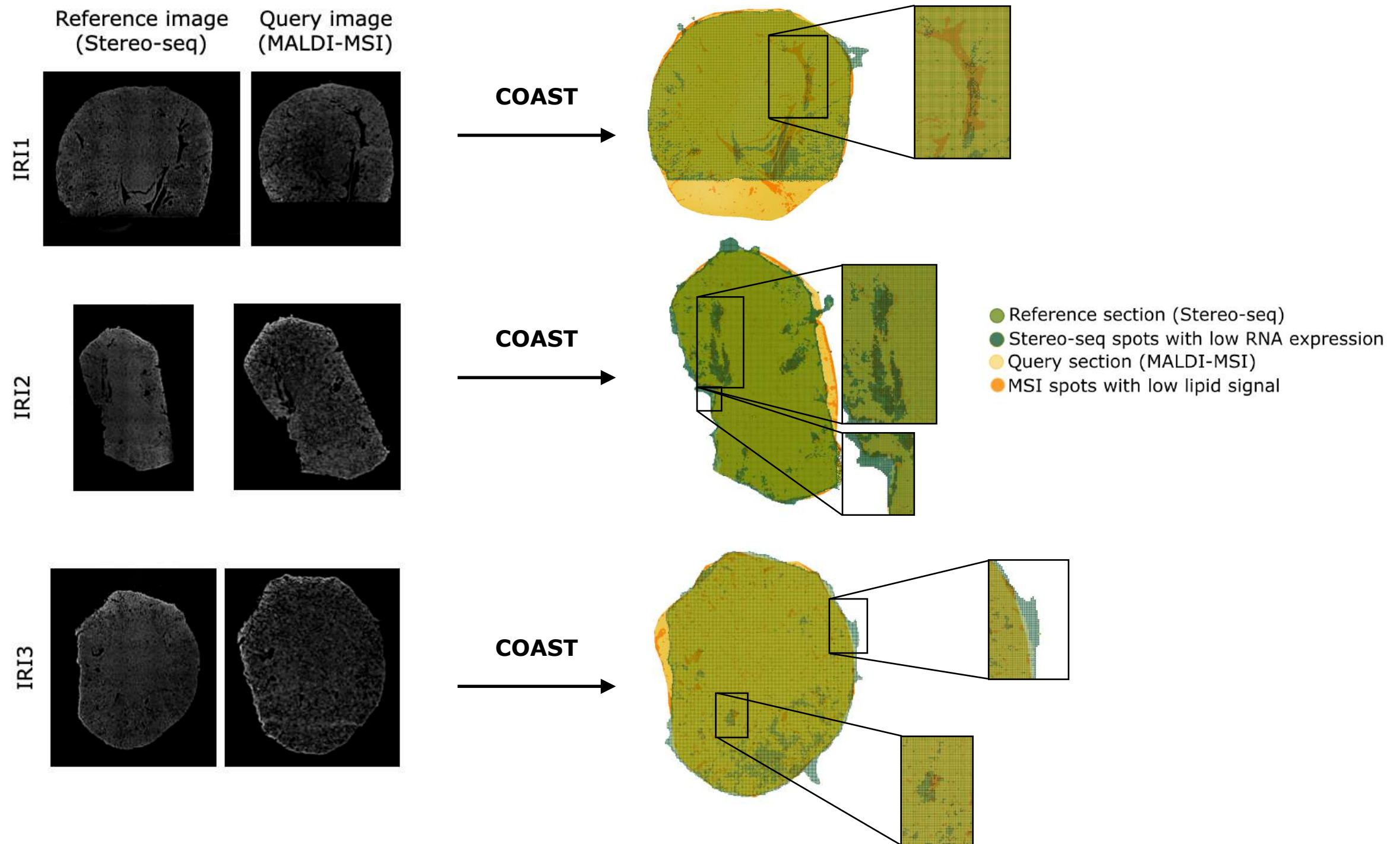


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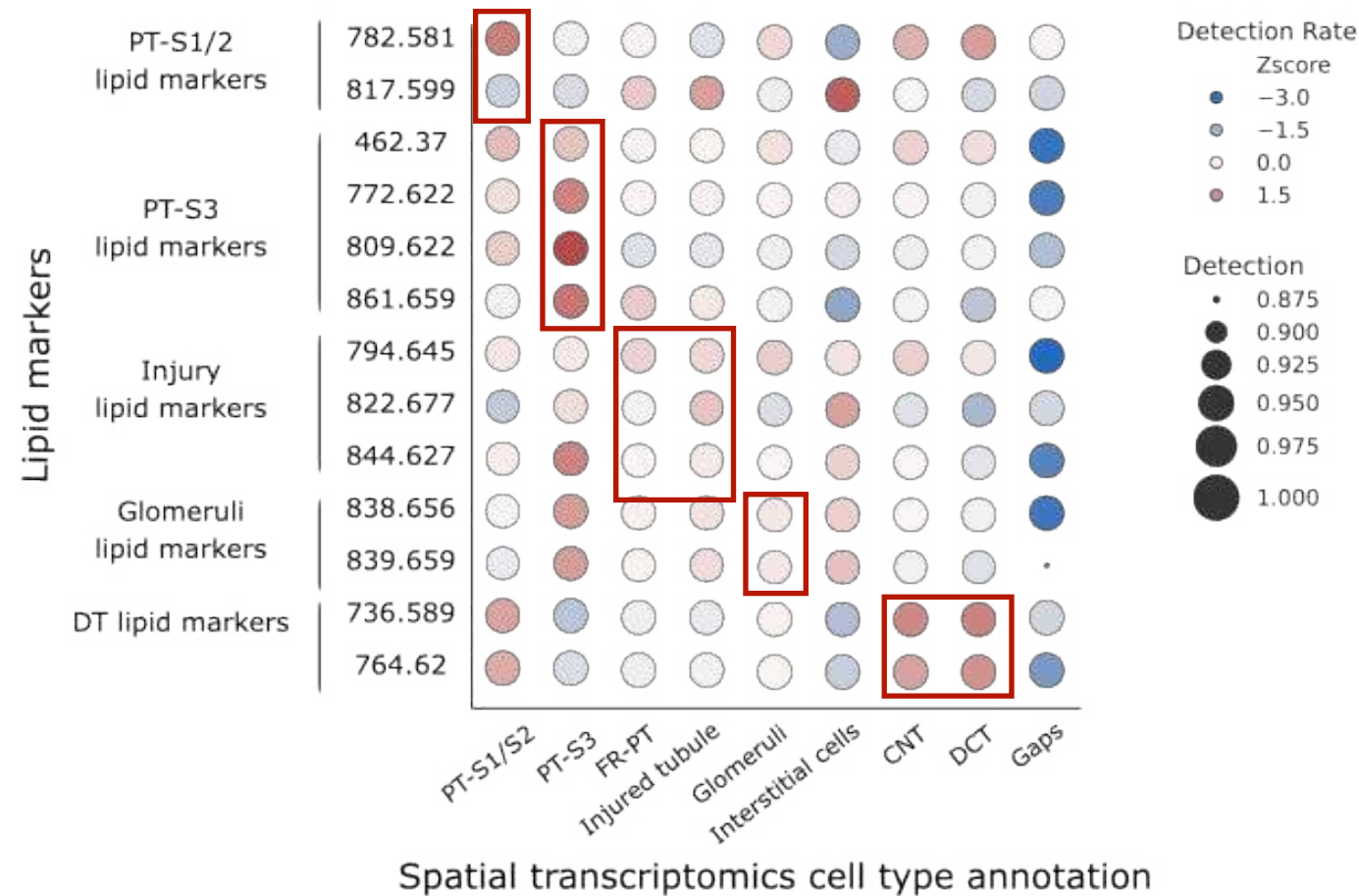
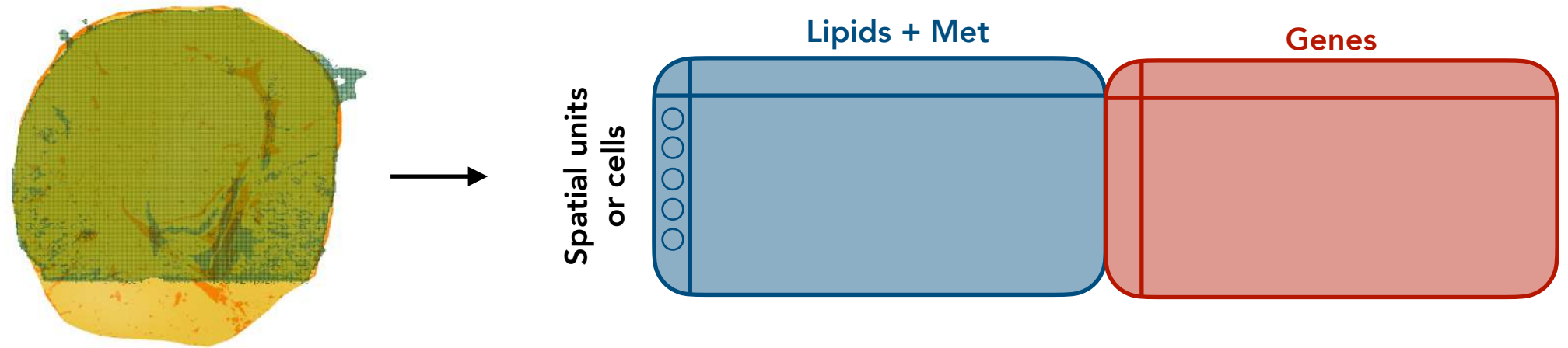


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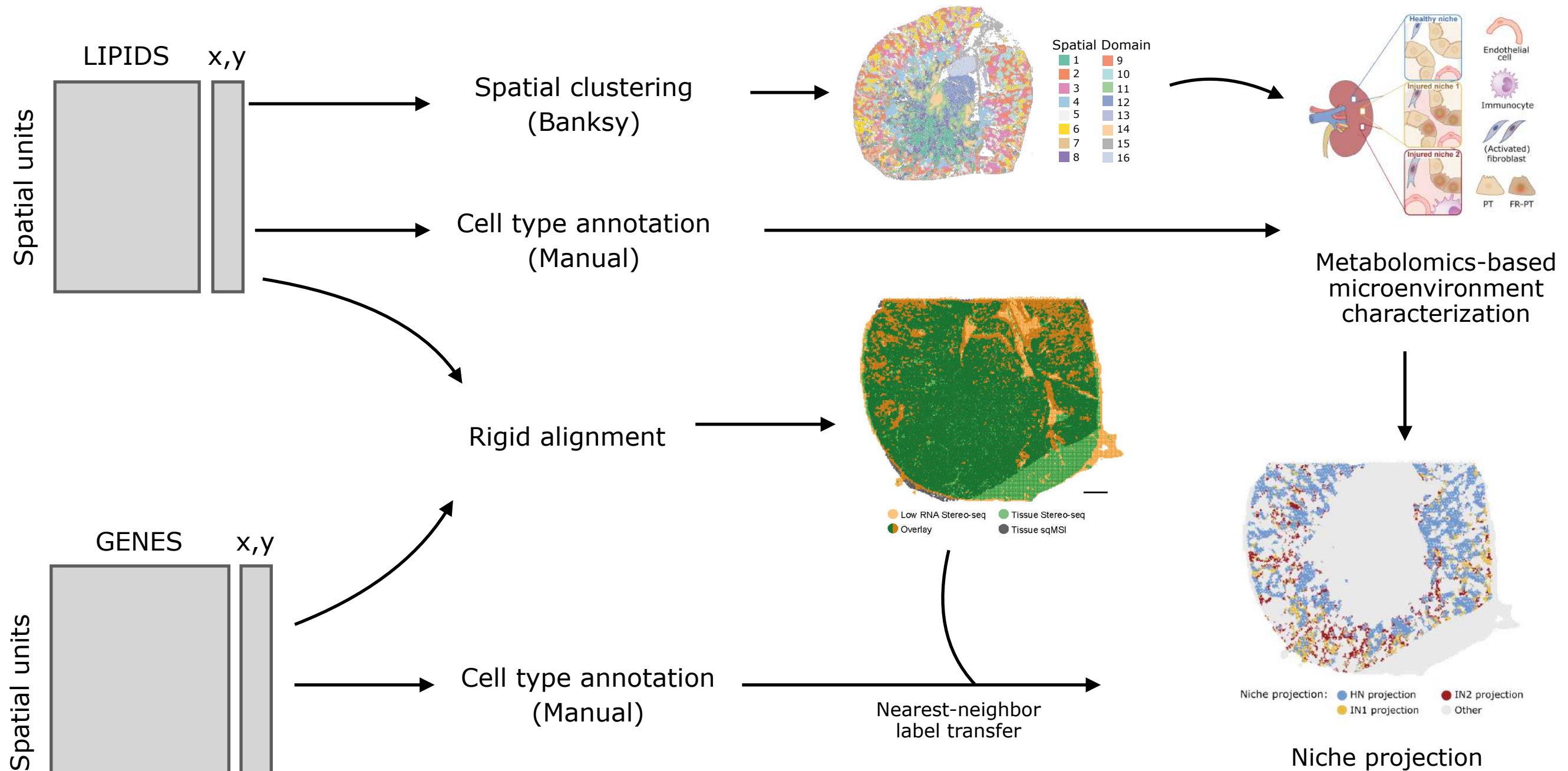
COAST can align multi-modal sections



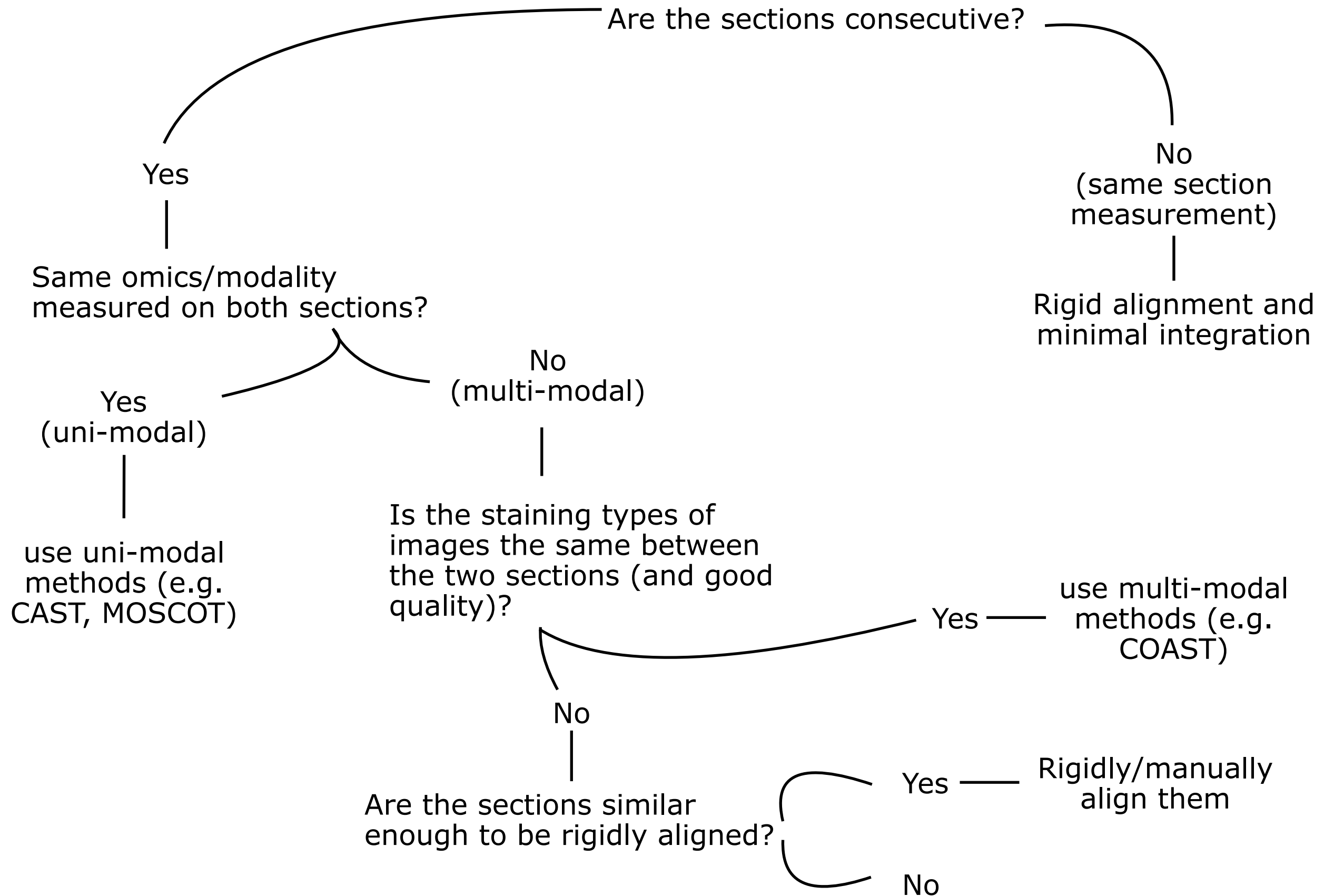
COAST can align multi-modal sections



Multi-modal study



Conclusions



References

1. Argelaguet, R., Cuomo, A.S.E., Stegle, O. *et al.* Computational principles and challenges in single-cell data integration. *Nat Biotechnol* **39**, 1202–1215 (2021). <https://doi.org/10.1038/s41587-021-00895-7>
2. Zhang, Meng, et al. “Molecularly Defined and Spatially Resolved Cell Atlas of the Whole Mouse Brain.” *Nature*, vol. 624, no. 7991, Dec. 2023, pp. 343–54, <https://doi.org/10.1038/s41586-023-06808-9>.
3. Chen, Ao, et al. “Spatiotemporal Transcriptomic Atlas of Mouse Organogenesis Using DNA Nanoball-Patterned Arrays.” *Cell*, vol. 185, no. 10, May 2022, pp. 1777–92.e21, <https://doi.org/10.1016/j.cell.2022.04.003>.
4. Fusar Bassini, L. et al. "The lipidomic architecture of the mouse brain". *Nature*. 2026. <https://doi.org/10.1038/s41586-026-11050-0>
5. Nunes, Joana B., et al. “Integration of Mass Cytometry and Mass Spectrometry Imaging for Spatially Resolved Single-Cell Metabolic Profiling.” *Nature Methods*, vol. 21, no. 10, Aug. 2024, pp. 1796–800, <https://doi.org/10.1038/s41592-024-02392-6>.
6. van Scheppingen, R. H., Wu, B., Zhang, Y., Schep, S., Winter, U., Wang, F., Zou, Y., Chuva de Sousa Lopes, S. M., Silver, D., Sidorov, I., Giera, M., van den Berg, B. M., van den Berg, C. W., Hui, L., Rabelink, T. J., & Wang, G. (2026). *Same-section spatial metabolo-transcriptomics using stereo-META-seq reveals DHA-driven kidney maturation*. <https://doi.org/10.64898/2026.06.16.732541>
7. Klein, Dominik, et al. “Mapping Cells through Time and Space with Moscot.” *Nature*, Nature Publishing Group, Jan. 2025, pp. 1–11, <https://doi.org/10.1038/s41586-024-08453-2>.
8. Tang, Z., Luo, S., Zeng, H. *et al.* Search and match across spatial omics samples at single-cell resolution. *Nature Methods* 21, 1818–1829 (2024).
9. Jones, Andrew, et al. “Alignment of Spatial Genomics Data Using Deep Gaussian Processes.” *Nature Methods*, vol. 20, no. 9, Aug. 2023, pp. 1379–87, <https://doi.org/10.1038/s41592-023-01972-2>.
10. Dosovitskiy, A., et al. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv:2010.11929 [Cs]*. <https://arxiv.org/abs/2010.11929>
11. Manzato, B., et al. “Image-Guided Alignment of Consecutive Multi-Modal Tissue Slides.” *Cell Reports Methods*, Sept. 2026, p. 101575, <https://doi.org/10.1016/j.crmeth.2026.101575>.